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Circular Economy in Beverage Packaging

A Case Study on the Innovation and Sustainability Potential of Refillable Glass Bottles

MOA AHLSTRAND

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KTH Industrial Engineering and Management
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SE-100 44 STOCKHOLM

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Cirkulär ekonomi inom dryckesförpackningar
En fallstudie om innovations- och hållbarhetspotentialen av återfyllningsbara glasflaskor

Moa Ahlstrand

Godkänt 2025-06-22	Examinator Sofia Ritzén	Handledare Johan Arekrans
	Uppdragsgivare Coca-Cola Europacific Partners	Kontaktperson Emma Blomborn

Sammanfattning

Avhandlingen undersöker potentialen för återfyllningsbara glasflaskor som en hållbar förpackningslösning för Coca-Cola Europacific Partners (CCEP) i Sverige, i linje med EU:s förordning för förpackningar och förpackningsavfall (PPWR). Studien utforskar hur återfyllningsbara förpackningssystem kan stödja principer för cirkulär ekonomi genom att minska miljöpåverkan, öka resurseffektiviteten och uppfylla hållbarhetsmål som satts av både myndigheter och företag. Metoden kombinerar semistrukturerade intervjuer med interna och externa intressenter, sekundärdataanalys och tematisk analys. Arbetet fokuserar på nyckelområden som regelverksefterlevnad, optimering av leveranskedjan, miljömässig hållbarhet, affärsstrategier och konsumentengagemang. Särskild uppmärksamhet riktas mot pantsystemet, som möjliggör effektiv insamling, sortering och återanvändning av glasflaskor, vilket säkerställer deras återintegrering i leveranskedjor och produktionsprocesser.

Resultaten belyser de operativa, tekniska och strategiska förändringar som krävs för att implementera ett framgångsrikt återfyllningsbart förpackningssystem. Viktiga rekommendationer inkluderar att utnyttja innovativa spårningsteknologier för att övervaka flaskornas livscykler för att effektivisera returflöden samt utforma insikter om initiativ för konsument- och kundengagemang för att maximera deltagandet i systemet. Livscykelanalys (LCA) visar de miljömässiga fördelarna med RGB system jämfört med engångsförpackningar, med betydande minskningar av koldioxidutsläpp, energiförbrukning och råmaterialanvändning över flera återanvändningscykler.

Studien bidrar till konkreta insikter och rekommendationer för CCEP och liknande organisationer som strävar efter att anpassa sig till föränderliga hållbarhetsregler och konsumentförväntningar. Genom att analysera olika perspektiv på hur innovativa förpackningslösningar, konsumentengagemang och regelverksefterlevnad kan integreras inom ramen för cirkulär ekonomi, stärks det akademiska bidraget. Undersökningen om utmaningar och möjligheter relaterade till leveranskedjor, innovativ teknik och konsumentbeteende erbjuder insikter för effektiva och hållbara förpackningslösningar inom dryckesindustrin. Resultaten betonar det återfyllningsbara förpackningssystemets centrala roll i att driva hållbar utveckling, möjliggöra innovativa lösningar och koppla ihop principer inom cirkulär ekonomi genom att minska avfallsproduktionen innan den når avfallshanteringsstadiet.

Nyckelord: Förpackningsinnovation, Återfyllningsbara glasflaskor, Cirkulär ekonomi, Avfallshantering, Hållbar förpackning, Pantsystem, Omvänd logistik, Sluten leveranskedja, Livscykelanalys, Konsumentengagemang, Regelverksefterlevnad, Ekologisk hållbarhet.

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Approved	Examiner	Supervisor
2025-06-22	Sofia Ritzén	Johan Arekrans
	Commissioner	Contact person
	Coca-Cola Europacific Partners	Emma Blomborn

Abstract

This thesis investigates the potential of refillable glass bottles (RGBs) as a sustainable packaging solution for Coca-Cola Europacific Partners (CCEP) in Sweden, in alignment with the EU Packaging and Packaging Waste Regulation (PPWR). The study examines how RGBs support circular economy principles by reducing environmental impacts, enhancing resource efficiency, and meeting sustainability goals set by both regulators and companies. A mixed-methods approach was utilized, combining semi-structured interviews with internal and external stakeholders, secondary data analysis, and thematic analysis. The research focuses on the key areas: regulatory compliance, supply chain, environmental sustainability, business strategy and engagement strategies. Special attention is given to the Deposit Return System (DRS) as an enabler for efficient collection, sorting, and reuse of glass bottles, ensuring their seamless reintegration into supply chains and production processes.

The findings highlight the operational, technological, and strategic changes required for implementing a successful RGB system. Key recommendations include leveraging innovative tracking technologies such as RFID and QR codes to monitor bottle lifecycles, optimizing reverse logistics to streamline return flows, and designing tailored consumer and customer engagement initiatives to maximize participation rates. Life Cycle Assessment (LCA) data demonstrates the environmental advantages of RGBs compared to single-use packaging, revealing significant reductions in carbon emissions, energy consumption, and raw material usage across multiple reuse cycles.

This study provides actionable insights for CCEP and other organizations striving to align with evolving sustainability regulations and consumer expectations. Additionally, it contributes to academic research by offering new perspectives on the integration of technological innovations, consumer engagement strategies, and regulatory compliance within the context of circular economy principles. By addressing challenges related to supply chains, technology adoption, and behavioral dynamics, this research offers a roadmap for scalable, effective, and sustainable packaging practices in the beverage industry. The findings underscore RGB systems role in advancing environmental stewardship, driving innovation, and fostering a circular economy and waste management.

Keywords: Packaging Innovation, Refillable Glass Bottles, Circular Economy, Waste Management, Sustainable Packaging, Deposit Return System, Reverse Logistics, Closed-Loop Supply Chain, Life Cycle Assessment, Consumer Engagement, Regulatory Compliance, Environmental Sustainability.

FOREWORD

I would like to express my deepest gratitude to my supervisor, Johan Arekrans, at KTH Royal Institute of Technology, for his insightful feedback, unwavering support, and invaluable guidance throughout this project. The provided expertise and mentorship were instrumental in navigating the complexities of this research and ensuring its academic contribution.

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I would also like to extend my heartfelt appreciation to the interview participants who generously shared their time, knowledge, and unique perspectives. Their valuable contributions provided critical insights that greatly enhanced the findings and outcomes of this thesis.

Finally, I am deeply grateful to my family and friends for their encouragement and belief in me during this challenging and rewarding journey.

Moa Ahlstrand

Stockholm, February 2025

NOMENCLATURE

Abbreviations	Description
CCEP	Coca-Cola Europacific Partners
CCEP DE	Coca-Cola Europacific Partners Germany
CCEP SWE	Coca-Cola Europacific Partners Sweden
CEE	Circular Economy Ecosystem
CLSC	Closed-Loop Supply Chain
CO_2	Carbon Dioxide
CSR	Corporate Social Responsibility
DRS	Deposit Return System/Scheme
FMCG	Fast-moving Consumer Goods
FSC	Forward Supply Chain
GHG	Greenhouse Gas
HoReCa	Hotel, Restaurant, Café Sector
HVO	Hydrotreated Vegetable Oil Fuel
LCA	Life Cycle Assessment
NRGB	Non-Refillable/Reusable Glass Bottle
PET	Polyethylene Terephthalate
PPWR	Packaging and Packaging Waste Regulation
QESH	Quality, Environment, Safety, and Health
RFID	Radio Frequency Identification
RGB	Refillable/Reusable Glass Bottle
rPET	Recycled Polyethylene Terephthalate
RSC	Reverse Supply Chain
RVM	Reverse Vending Machine
TCCC	The Coca-Cola Company

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1 INTRODUCTION

This chapter presents the background, purpose, limitations, and methodology of the project, detailing Coca-Cola Europacific Partners' (CCEP) efforts towards sustainability within the Swedish market regarding regulatory frameworks and innovative packaging solutions. The chapter provides an overview of CCEP's operational strategy, sustainability initiatives, and the role of refillable packaging in achieving circular economy goals.

1.1 Background

Coca-Cola Europacific Partners (CCEP) the one of the largest bottling partners of The Coca-Cola Company (TCCC), producing and distributing beverages across Europe, Australia, and parts of Asia (Coca-Cola Europacific Partners, 2023a; The Coca-Cola Company, 2024). As part of the Coca-Cola system, CCEP operates under a model partnership, sourcing, production, and recycling to distribute products through various channels.

In Jordbro, Sweden, CCEP operates a production facility employing 725 staff and contributing to 2.0% of the total European revenue (Coca-Cola Europacific Partners, 2023a). Recent projects in Sweden, including expanded warehouse capacity and the use of electric trucks, emphasize their commitment to operational efficiency and environmental impact reduction (Coca-Cola Europacific Partners, 2023b).

CCEP's sustainability strategy, "This is Forward" launched in 2017, aligns with TCCC's global goals. It focuses on areas such as climate action, sustainable packaging, and waste reduction, which are in line with the UN Sustainable Development Goals (Coca-Cola Europacific Partners, 2023d). Furthermore, the "World Without Waste" initiative from TCCC, targets a significant reduction in single-use packaging by 2030, aiming for at least 25% of global beverage volume in refillable or returnable formats (The Coca-Cola Company, 2023).

The recent EU Packaging and Packaging Waste Regulation (PPWR) is expected to significantly impact the beverage industry (Council of the European Union, 2024). The PPWR is aimed to enhance sustainability and reduce packaging waste across the EU. A key objective of the PPWR is to increase the use of reusable packaging systems within the beverage sector. By January 1, 2030, distributors of alcoholic and non-alcoholic beverages within the EU should ensure that at least 10% of their products are available in "reusable packaging within a system for re-use or by enabling refill", with this target rising to 40% by January 1, 2040. These targets are designed to align with the principles of a circular economy, aiming to minimize waste through repeated reuse cycles and encouraging the more sustainable use of materials.

The regulation also highlights the role of closed-loop systems, where reusable packaging stays under consistent ownership throughout its lifecycle, including stages of collection, reconditioning, and redistribution (Council of the European Union, 2024). Implementing such systems necessitates robust reverse logistics to collect and prepare reusable packaging for subsequent use. An integral part of this process is the implementation of Deposit Return Systems (DRS), which incentivize consumers to return bottles for reuse or high-quality recycling. DRS plays a crucial role in meeting the PPWR's objectives by ensuring that bottles reuse system are efficiently collected, processed, and returned to the market.

Additionally, the PPWR mandates that packaging designs prioritize sustainability, requiring high recyclability and the minimal use of harmful substances (Council of the European Union, 2024). This includes designing bottles that can withstand multiple reuse cycles while maintaining safety,

quality, and hygiene standards. The regulation's holistic approach aims to foster innovation in packaging and ensure that these solutions support the EU's long-term environmental goals. In summary, these new requirements present significant challenges and opportunities for companies such as CCEP.

1.2 Purpose

This study focuses on the implementation of refillable glass bottles (RGBs) as a strategic response to the sustainability goals set by Coca-Cola Europacific Partners (CCEP) and The Coca-Cola Company (TCCC), aligned with the requirements of the EU's Packaging and Packaging Waste Regulation (PPWR). To prepare for future regulatory landscapes, the study explores how these regulations drive the adoption of reusable packaging systems, analyzing the role of Deposit Return Systems (DRS) and reverse logistics with an innovative approach. These processes are essential for establishing a closed-loop system, ensuring that bottles are efficiently returned, processed, and reintegrated into the supply chain.

The study evaluates CCEP's strategies to meet these regulatory requirements, focusing on the operational and logistical adjustments to create a system for RGBs. It considers the consumer perspective, investigating how to engage customers in DRS and encourage positive recycling behaviors, as consumer participation is a critical element of the RGB system. The study also explores the use of innovative technology to enhance consumer engagement and streamline the return process, making the system more efficient and customer centric.

Additionally, the study analyzes sustainability metrics, using life cycle thinking. This involves assessing the environmental impact of RGBs throughout their entire lifecycle providing insights into how the implementation affects overall carbon emissions and aligns with CCEP's sustainability objectives. It examines how DRS can be optimized to facilitate the return of bottles from consumers back into the supply chain, ensuring a seamless cycle of reuse while minimizing environmental impact.

The study further explores larger-scale implications of its findings, connecting the results to the strategic sustainability goals of CCEP and TCCC within the beverage industry. This broader analysis offers insights into how compliance with regulations is driving a transformative shift toward more sustainable and circular packaging solutions. By providing a comprehensive view of the impact of these changes, the study aims to position CCEP and TCCC to maintain their leadership in sustainable packaging practices, ensuring they remain at the forefront of industry innovation within an evolving regulatory landscape.

1.3 Research Questions

RQ1: What strategies can be employed to maximize return rates of bottles in a RGB system?

This research question focuses on identifying and analyzing effective methods for increasing customer and consumer participation in return systems, particularly in DRS and reverse logistics. The analysis will cover the following key areas:

1. **Behavior and Engagement:**
   Investigating how to effectively engage customers and customers in the return process and encourage positive recycling behaviors. Consumer participation is a critical element of the RGB system, and this section will analyze customer and consumer engagement strategies.

2. **Innovative Technologies:**
   Exploring how innovative technologies can enhance a RGB system, customer engagement, streamline the return process, and improve the efficiency of reverse logistics, making the system more customer-centric and effective.

3. **Supply Chain:**
   Analyzing the key adjustments needed to develop a system for RGBs. This section focuses on the challenges of maintaining a closed-loop system that ensures bottles are efficiently returned, processed, and reintegrated into the supply chain, with a view toward potential future improvements.

RQ2: What are the strategic connections of implementing a RGB system for CCEP?

This question broadens the scope of the study and contextualize it by examining how implementing RGBs aligns with the broader goals of CCEP. It will explore the following areas:

1. **Regulatory Compliance:**
   Assessing how the strategies align with the requirements of the EU's Packaging and Packaging Waste Regulation (PPWR) and other relevant laws, investigating how the return system is compliant with current and future regulations.

2. **Environmental Sustainability:**
   Using life cycle thinking to evaluate the environmental impact of RGBs across their lifecycle. This section will analyze how these strategies impact overall carbon emissions and how they align with CCEP's sustainability objectives.

3. **Strategic and Business Objectives:**
   Evaluating the broader implications of these strategies for CCEP's leadership in the beverage industry. This involves analyzing the transformative shift with RGBs toward a more sustainable and circular packaging solutions would be seen as strategically with a focus on business, innovation and sustainability.

1.4 Research Gap and Contributions

This study provides an in-depth analysis of the practical application of circular economy principles within the beverage industry, with a focus on RGBs. It advances knowledge by demonstrating how regulatory frameworks, such as the EU's Packaging and Packaging Waste Regulation (PPWR), can be integrated with corporate sustainability strategies. The research highlights how large-scale companies can adapt to and benefit from these regulations, offering a roadmap for aligning business operations with environmental standards.

The study also contributes significantly to the field of sustainable logistics by examining the role of consumer behavior and technology in optimizing reverse logistics systems and improving return rates. It offers practical recommendations for operational changes required to establish efficient DRS and emphasizes how existing Life Cycle Assessment (LCA) data can be leveraged for strategic environmental decision-making.

Addressing a gap in existing literature, this research links current and emerging regulatory frameworks, corporate sustainability strategies, and consumer engagement. It provides valuable insights into how CCEP and similar companies can align their operations with evolving regulations and sustainability goals. The findings serve as a reference for future studies on sustainable

packaging systems and offer practical guidance for businesses looking to enhance their circular practices in response to changing environmental regulations and consumer expectations.

By bridging regulatory compliance with strategic sustainability initiatives, this study presents a comprehensive perspective that can guide future research and assist companies in fostering sustainable and circular packaging solutions in the beverage industry.

1.5 Delimitations

Certain areas are excluded from the scope of this research. The broader supply chain of CCEP outside Sweden, detailed financial analyzes of transition costs, and comparisons with non-European regulatory frameworks are not examined where the regulatory focus remains on the PPWR. While LCAs are used to evaluate environmental impacts, this study relies on existing LCA data instead of conducting new assessments.

This study does not include detailed technical assessments of production processes, which would involve comprehensive analysis of engineering, manufacturing, and technological aspects related to producing and handling RGBs. Such assessments could encompass evaluations of machinery, modifications to production lines, and feasibility studies of specific processes within the factory environment.

Similarly, in-depth business analyzes are beyond the scope of this research. These strategies would be related to financial planning, investment evaluations, such as detailed cost-benefit analyzes of transitioning to refillable systems and budgeting considerations.

2 METHODOLOGY

This chapter outlines the research method used to gather quantitative and qualitative data regarding the analysis of a RGB system at CCEP. The methodology is designed to provide a comprehensive understanding of the challenges involved in transitioning from Non-Refillable Glass Bottles (NRGBs) to Refillable Glass Bottles (RGBs). The research approach integrates semi-structured interviews, quantitative data collection, and regulatory analysis to develop actionable insights for a sustainable and efficient refillable bottle system.

Mixed methods research combines qualitative and quantitative approaches to provide a comprehensive understanding and corroboration of findings and answer the research questions (Schoonenboom and Johnson, 2017). As described, mixed methods designs can range from simple to complex, with integration occurring at various stages, such as data collection, analysis, or interpretation. The study adopted a mixed-methods research design to gain a comprehensive understanding of the strategies and challenges involved in implementing a system for RGBs at CCEP, analyzing qualitative and quantitative data in parallel. The approach followed principles from the source, emphasizing iterative refinement to align goals, questions, and methods. This flexibility allowed the methodology to adapt as new insights emerged, ensuring a robust and comprehensive analysis to answer the research questions, The integration of qualitative and quantitative data resulted in a more comprehensive understanding than either approach could achieve independently (Proudfoot, 2023).

2.1 Literature Review

The literature review established the theoretical foundation for this study by focusing on RGBs, DRS, reverse logistics, and sustainability metrics. A comprehensive search strategy was employed to ensure a robust contextual background for the analysis. Web of Science was chosen as the primary database due to its extensive collection of high-quality, peer-reviewed journal articles, providing credible insights into sustainability, logistics, and circular economy principles. The search process employed targeted keywords, such as "Refillable Glass Bottle," "Reverse Vending Machines," "Glass Deposit System," "Closed Loop Glass Bottle," "Circular Economy," and "Tracking Technology Glass Bottles." These keywords were carefully chosen to capture the study's core themes while focusing on recent developments in refillable packaging systems. This ensured that the analysis was firmly grounded in both academic research and current industry practices. To complement this academic research, other relevant information was gathered using Google, sourcing sustainability reports, corporate publications, and industry data. This dual approach ensured a balanced understanding of both theoretical frameworks and practical applications.

This study employed a systematic literature review to synthesize prior research, evaluate existing knowledge, and identify gaps, following the guidelines established Xiao and Watson (Xiao and Watson, 2019). Systematic literature reviews can be categorized into four types: descriptive, which maps the current state of the literature and highlights gaps; testing, which evaluates hypotheses or questions using qualitative or quantitative methods; extending, which develops new theories or frameworks from existing literature; and critical, which assesses the quality and rigor of previous studies. The approach taken in this study incorporates mainly the descriptive review to map out the current state of the scope by organizing relevant studies to identify patterns and gaps. Articles were prioritized based on their relevance to key topics such as RGB and DRS systems and consumer engagement. Preference was given to studies with a European or Swedish context to align with the regulatory and operational focus of this research. Recent publications were emphasized to capture the latest industry trends, regulatory developments, and sustainability

innovations, including the PPWR. Credibility was ensured by focusing on journal articles and reputable industry reports.

This methodical approach to sourcing and selecting literature provided a strong foundation for the study, bridging the gap between theory and practice. By integrating insights from academic research and industry data, the literature review supports a comprehensive analysis of how RGB systems align with CCEP's sustainability goals and broader strategic objectives, while addressing regulatory and operational challenges.

2.2 Qualitative Data

Phase 1: Internal

The first phase of the research involved conducting semi-structured interviews with key stakeholders and experts at CCEP. Semi-structured interviews were chosen to allow flexibility in exploring specific topics relevant to each stakeholder's area of expertise while ensuring a consistent set of core topics across all interviews. The semi-structured interview is valued for its flexibility and ability to foster meaningful, reciprocal engagement between interviewer and participant (Kallio *et al.*, 2016). This method was particularly useful in uncovering the operational nuances and strategic priorities that differ between departments but are critical to the overall implementation of an RGB system.

These interviews were designed to gather in-depth insights into the challenges and opportunities in implementing a RGB system, focusing on customer and operational logistics, sustainability, innovation, production/manufacturing and regulatory compliance at country and EU levels. In common with all the interviews was the focus on RGBs but the topics discussed could vary depending on the expert area of the interviewee.

The stakeholders represented various areas such as manufacturing, logistics, sustainability, QESH (Quality, Environment, Safety and Health), and cross-country wise to get a holistic approach shown in Table 1.

Table 1. Interviews Phase 1

Title/Role	Company	Duration	Type of Interview	Key Topics Discussed	Documentation Method
Sr Manager, Central QESH European Supply Chain	CCEP	60 min	Virtual	Sustainability, quality, supply chain	Took notes, recorded session
Sr Manager, Distribution Operations	CCEP	50 min	Virtual	Operations of glass production line, distribution	Took notes, recorded session
Manager, Environment & Energy	CCEP	30 min	Virtual	CCEP Germany perspective on RGB	Took notes, recorded session
Manager, Field Environment	CCEP	30 min	Virtual	CCEP Germany perspective on RGB	Took notes, recorded session
Sr Manager, Manufacturing Transformation SME - Capital Projects CE	CCEP	120 min	In-person	Manufacturing, operations	Took notes
Sr Manager, Customer Logistics	CCEP	45 min	In-person	Customer logistics	Took notes

Valuable insights regarding key operational aspects of the current glass bottle system emerged from the interviews. These insights, combined with the secondary data, allowed for a modeling of the existing glass bottle line. This model provided an understanding of the logistical flows.

Additionally, this phase played a crucial role in refining the study's scope and research questions. The exploratory nature of the interviews allowed for iterative development of the study's focus, aligning it with both operational realities and strategic priorities.

From these interviews, valuable insights emerged regarding key operational aspects of the current glass bottle system. These insights, combined with secondary data, facilitated the development of a model of the RGB closed-loop system and its production activities. RGB system includes all interconnected processes, including production, logistics, collection, and reconditioning of bottles. This model provided a comprehensive understanding of logistical flows, highlighting areas for improvement and optimization.

Moreover, this phase was important in identifying gaps and refining the research questions. The exploratory nature of the interviews enabled the recognition of areas requiring deeper investigation, allowing for iterative development of the study's focus. This ensured that the research aligned closely with operational realities while addressing strategic priorities and uncovering key opportunities for further analysis.

Phase 2: Stakeholder Interviews and Reverse Logistics

In the second phase, semi-structured interviews were conducted with stakeholders at CCEP, TCCC, Sveriges Bryggerier and Systembolaget. This phase specifically focused on reverse logistics and DRS. The aim was to gain insights challenges and opportunities of RGBs, focusing on customer and consumer perspective, innovative technologies in packaging, regulatory compliance and strategic alignment.

The stakeholders, representing various areas such as packaging, operations, customer knowledge and current DRS and waste management in Sweden, shown in Table 2. The goal was to gather knowledge in these areas to find gaps to develop strategies to maximize return rates from customer in a RGB line, investigate the innovative tracking solution, analyze alignment with the PPWR and investigate the strategic objectives of a sustainable shift with RGBs.

Table 2. Interviews Phase 2

Title/Role	Company	Duration	Type of Interview	Key Topics Discussed	Documentation Method
Director, Packaging Development Coca-Cola Global Development & Innovation EMEA	TCCC	1h	Virtual	Packaging and sustainability, strategy	Took notes, recorded session
Senior Director, Technical Operations	TCCC	1h	Virtual	Technical operations and packaging innovations	Took notes, recorded session
Senior Manager, Technical Packaging	CCEP	1h	Virtual	Packaging innovation	Took notes, recorded session
Associate Director, Channel Development & Shopper Activation	CCEP	1h	Virtual	Customer marketing and reverse logistics, HOME customer segment	Took notes, recorded session
Head of Returglas, Production, Logistics & Quality	Sveriges Bryggerier	1h	Virtual	Consumer behavior and reverse logistics, RGBs	Took notes, recorded session
Manager, Field Environment	CCEP	30 min	Virtual	Germany’s perspective on RGB	Took notes, recorded session

Sr Manager, Equipment Development	CCEP	1h	Virtual	Away-From-Home customer segment, pilot projects	Took notes, recorded session
Doctoral Researcher, Chemical Engineer - LCA and Energy Storage Systems	KTH	1h	Virtual	LCA thinking	Took notes, recorded session
Director, Franchise Operations	TCCC	1h	Virtual	Business strategy, innovation & sustainability	Took notes, recorded session
Area Manager, Sustainability & Communication	Systembol aget AB	1h	Virtual	Customer perspective, RGB system, sustainability	Took notes, recorded session
Customer Manager	Returpack	1h	Virtual	DRS operations, consumer engagement strategies	Took notes, recorded session

These interviews provided valuable insights into consumer behavior, the potential for innovative technologies to improve return rates, and the challenges involved in implementing an effective reverse logistics system. The interviews also explored the alignment of these technologies and strategies with current and regulatory requirements and strategical alignment.

2.3 Secondary Data

In addition to the qualitative interviews, quantitative data was gathered to support the sustainability analysis of the RGB systems. Key data sources included sustainability metrics, LCA, and performance data. A multivocal literature review combines peer-reviewed academic sources and grey to capture both research and industry perspectives according to a study (Garousi *et al.*, 2019). Gray literature is defined as material that is not controlled by commercial publishers or published in academic journals and offers the benefit of contributing to unique perspectives, avoiding publication bias, and providing practical and contextual insights. This method is valuable as it fills gaps by academic publications and could offer insights into emerging trends. The usage of both types of literature in this report is connected to the theory of multivocal literature review which involve three phases: planning, where research questions and inclusion criteria are defined; conducting, which includes searching for and assessing both academic and GL sources; and reporting, where findings are documented transparently. Reports from CCEP were used to provide a comprehensive understanding of the performance of the glass bottle systems. These reports offered detailed insights into logistics, production, manufacturing, sales volumes, and customer behavior. This information was utilized to evaluate the current system's performance and to identify potential areas for improvement, ensuring the research was grounded in practical, real-world data.

The Life Cycle Assessment (LCA) conducted by the Research Institute of Sweden (RISE) in collaboration with Sveriges Bryggerier was central to the overall analysis in this master's thesis, playing a key role in addressing the research questions. This LCA provided invaluable insights and into the environmental impact of single-use and RGB systems, focusing on factors such as carbon footprint. The LCA data served as the base for modeling the potential sustainability benefits of an RGB system, particularly its ability to reduce CO_2 emissions. These findings enabled a comparative analysis between the current glass bottle system and the proposed RGB system.

A comparative analysis of Coca-Cola Europacific Partners Germany (CCEP DE) and Coca-Cola Europacific Partners Sweden (CCEP SWE) was conducted by reviewing reports. The investigation

also considered cultural factors and operational scale to highlight challenges and opportunities for RGB systems in Sweden.

Regulatory frameworks, such as the EU's Packaging and Packaging Waste Regulation (PPWR) and national guidelines, were also incorporated into the secondary data collection. These regulations provided critical parameters for ensuring that the proposed RGB system would comply with existing and future sustainability standards. The regulatory data was used to align the strategic recommendations with both environmental goals and compliance requirements.

Data analysis also included a focused review of reverse logistics, emphasizing the existing RGB infrastructure. This encompassed an evaluation of Reverse Vending Machines and logistical challenges specific to urban centers. The analysis aimed to identify the effectiveness of current return mechanisms and suggest enhancements to optimize return rates and system efficiency. By examining these elements, the study provided actionable insights for refining reverse logistics processes and aligning them with sustainability goals. Further data was collected on the current DRS in Sweden, which provided additional context on how the RGB system could be integrated into the existing return infrastructure. Insights from the DRS helped identify opportunities for optimizing return rates, logistics, and customer engagement strategies to maximize the effectiveness of the RGB system within the Swedish market.

2.4 Thematic Analysis

A thematic analysis is a qualitative method for identifying, analyzing, and reporting patterns in data (Kiger and Varpio, 2020). It is flexible and applicable allowing to explore both explicit and deeper meanings. The data analysis employed a combination of inductive and deductive approaches to ensure a comprehensive understanding of themes related to the RGB system.

Deductive thematic analysis used predefined theoretical categories from existing literature to ensure alignment and rigor, utilizing existing literature and theoretical frameworks to identify initial themes (Proudfoot, 2023). In contrast, the inductive approach followed key steps such as data familiarization, coding, theme identification, theme review, defining and refining themes, and concluding the final analysis, allowing new insights to emerge naturally from the data, capturing unanticipated findings.

Data from interviews were analyzed using thematic analysis to identify patterns and themes related to supply chain challenges, strategic alignment, innovative technology, customer and consumer behaviors, and compliance strategies and connecting it to theoretical foundation gathered from the literature review. The interviews were transcribed and analyzed to extract codes and generate themes. After initial theme development, a review was conducted to refine these themes, revisiting the transcriptions to find supporting quotes and ensure nothing was overlooked. The main themes were divided into further subthemes reflecting on the recommendations in the relevant areas regarding RGBs in Sweden. This strategy provided a multi-layered understanding of the data.

2.5 Digital Tools

Throughout the study, a variety of digital tools were utilized to enhance data collection, analysis, visualization, and brainstorming processes. ChatGPT served as a collaborative platform for brainstorming ideas, performing initial data analysis, and compiling information in a structured manner. This tool facilitated the refinement of complex concepts and supported the analysis.

Microsoft Teams was employed for recording and transcribing interviews, providing detailed and accurate records that streamlined the review and coding process for thematic analysis. Figma played a critical role in visualizing the RGB and NRGB systems. This tool was particularly useful for visual brainstorming and mapping out ideas, aiding in the initial stages of conceptualization and continuing to support visual representation throughout the project.

Excel was used extensively for organizing, coding, and analyzing both qualitative and quantitative data, enabling the categorization and visualization of themes that informed targeted recommendations. It was also used to calculate different types of data. Word was essential for drafting and structuring the written sections of the thesis, ensuring clarity and coherence in presenting findings.

The strategic use of these digital tools ensured a streamlined, comprehensive, and organized approach throughout the study, contributing to the development of well-informed insights and recommendations.

2.6 Validity and Reliability

To ensure the reliability of the research, data from multiple sources, including interviews and company reports, were cross-referenced. This approach helped verify that findings were comprehensive and supported by diverse perspectives. Consistency was confirmed by comparing insights from different stakeholders to identify common patterns and validate results.

Interviewees were strategically chosen from various roles and areas of expertise to provide a holistic view of the challenges and opportunities associated with implementing the RGB system. This comprehensive selection strengthened the reliability of the study by incorporating a range of operational, strategic, and technical perspectives.

The inductive/deductive hybrid thematic analysis enhances rigor but shares limitations with thematic analysis, such as bias and replicability by (Proudfoot, 2023). On the other hand, a thematic analysis is well-suited for mixed-methods research for integrating qualitative and quantitative methods combined with theory-driven quantitative analysis within a critical realist framework.

Triangulation as a research method that involves examining an issue from at least two perspectives, focusing on integrating diverse viewpoints to enhance the robustness and validity of research findings (Archibald, 2016). The research employed methodological triangulation by integrating qualitative and quantitative methods. Additionally, data triangulation was used by drawing on various sources, including reports, external benchmarks, and literature reviews, to validate and enrich the analysis. This approach ensured balanced, well-supported insights and informed practical recommendations for enhancing the sustainability and efficiency of the RGB system.

3 CASE COMPANY

This chapter provides an overview of Coca-Cola Europacific Partners (CCEP), a key bottling partner within The Coca-Cola Company (TCCC) system. It examines CCEP's business model, sustainability initiatives, and market drivers, with a focus on its transition toward sustainable packaging, including Refillable Glass Bottles (RGBs). Key regional efforts, such as those in Sweden and Germany, are highlighted to showcase how CCEP integrates innovation and sustainability to address industry challenges and drive a circular economy.

3.1 Overview of Company

Coca-Cola Europacific Partners (CCEP) is recognized as one of the world's largest producers and distributors of beverage in Western Europe, Australia, Pacific, Indonesia and the Philippines (Coca-Cola Europacific Partners, 2023a) (The Coca-Cola Company, 2024). CCEP is one of The Coca-Cola Company's (TCCC) bottling partners, also known as bottlers, and are known together as the Coca-Cola system. CCEP's business model is based on the close relationship with TCCC and franchisors where they purchase concentrates, beverage base and syrups to make sell and distribute finished products to customers and vending partners (Coca-Cola Europacific Partners, 2023b). This collaborative process is captured in their "partner, source, make, recycle, sell and distribute" approach, emphasizing sustainability and operational integration.

TCCC has approximately 200 bottling partners and 950 production facilities worldwide, where the distribution is 2,2 billion servings per day (The Coca-Cola Company, 2024a). TCCC does not own, manage or control most of the bottlers and generate net operating revenue from the sales of the syrups and concentrates sold to authorized bottling partners (The Coca-Cola Company, n.d.). The finished product is made by the bottling partner where they combine concentrates with sweeteners, still or sparkling water, prepare packages and sell the beverage.

The company's sustainability strategy, known as "This is Forward" was introduced in 2017, and is integral to the long-term business model and aligned with TCCC's global sustainability mission (Coca-Cola Europacific Partners, 2023c). "This is Forward" centers around six key areas: climate action, sustainable packaging, water stewardship, promoting the well-being of its people and those across the value chain, offering consumers healthier choices, and contributing to local communities. These are connected to the United Nations Sustainable Development Goals (SDGs). The "World Without Waste" strategy includes design, collect and partner for TCCC and their partners and bottlers (The Coca-Cola Company, 2023). By 2030, the goal is for at least 25% of TCCC's global beverage volume to be sold in refillable or returnable glass and plastic bottles, or through fountain dispensers that use reusable packaging.

As a key partner of TCCC, Coca-Cola Europacific Partners (CCEP) aligns its actions with TCCC's strategies, such as the "World Without Waste" initiative, to achieve net-zero GHG emissions by 2040 (Coca-Cola Europacific Partners, 2023c). "Action on packaging" is a way to reduce environmental impact and to reach the goal for a net zero GHG emission goal (Coca-Cola Europacific Partners, 2021). In 2020, 84% of CCEP's total glass bottles were RGBs, further showcasing their efforts in sustainable innovation and packaging solutions.

3.2 Market Drivers

CCEP is influenced by market trends and macroeconomic factors, with sustainability emerging as a key driver (Coca-Cola Europacific Partners, 2023b). Government regulations on climate change and packaging, along with growing eco-consciousness among younger consumers, have prompted CCEP to prioritize environmental initiatives. To support these goals, CCEP established CCEP Ventures, a fund dedicated to advancing sustainable practices.

CCEP actively works to reduce emissions from its third-party logistics, which account for around 7% of Scope 3 GHG emissions (Coca-Cola Europacific Partners, 2023b). Through warehouse capacity optimization, efficient transportation with suppliers, and increased use of alternative fuels (currently 15% of total kilometers in Europe), CCEP lowers its CO_2 footprint. In regions like Belgium, Luxembourg, Spain, and Sweden, electric trucks are used for local deliveries to further cut emissions.

To minimize transport distances, CCEP partners with suppliers strategically, with some production sites, like Grigny and Halle, producing PET pre-forms in-house to streamline the supply chain (Coca-Cola Europacific Partners, 2023b). Through front- and back-hauling programs, CCEP ensures trucks are fully utilized in both directions, reducing empty miles and boosting efficiency and sustainability.

Technology is also a crucial market driver as consumers increasingly turn to digital channels (Coca-Cola Europacific Partners, 2023b). CCEP has embedded digital capabilities organization-wide, enabling innovation that streamlines logistics, packaging, and resource management, further enhancing sustainability efforts across its operations.

3.3 Coca-Cola Europacific Partners in Sweden

In 2023 Sweden had one production facility in Jordbro industrial area (Coca-Cola Europacific Partners, 2023d). With a total of 725 employees, it stood for 2,0% of the revenue in Europe. Total revenue was 2023 398 € million (Coca-Cola Europacific Partners, 2023a). In 2023 an external warehouse was opening to increase storage capacity reducing CO_2 footprint and costs while having flexible and efficient operations and minimizing truck movements (Coca-Cola Europacific Partners, 2023d). The deliveries to local customers are by electric trucks.

As a Fast-Moving Consumer Goods (FMCG) company the customers are important. According to a Gradient CSAT report 2023, CCEP was ranked as number one supplier withing beverage and FMCG in Sweden, and was number one in Belux, Great Britain, the Netherlands, Portugal, Spain which is six out of eight surveyed markets (Coca-Cola Europacific Partners, 2023d).

CCEP's dedication to innovation in sustainability is exemplified by Sweden was the first market worldwide where all plastic bottles were produced from 100% recycled material (The Coca-Cola Company, 2019). CCEP offers reusable/refillable PET (rPET) bottles across eight markets for locally produced beverage portfolios (The Coca-Cola Company, 2022). This transition, which began in 2020, has eliminated the use of 3,500 tonnes of virgin plastic annually and resulted in a 25% reduction in CO_2 emissions over pre-transition levels. This highlights CCEP's willingness to adopt innovative approaches to leading in circular economy practices and aligns with their 'World Without Waste' initiative to recover and reuse packaging material and meet the demand for sustainable packaging.

In 2023 6.9% of the volume was through dispensed solutions where a single use to reusable drinking solutions was encouraged (Coca-Cola Europacific Partners, 2023d). In Sweden a partner with a large fast-food chain, Burger King to test dine-in reusable cups to see the potential of

reduced waste and GHG emissions. Furthermore, pilot project in Sweden was tested to meet consumer demands where innovations were implemented (Coca-Cola Europacific Partners, 2023c)

3.4 Cross-Country Efforts in Refillable Beverage Packaging

As part of its efforts to increase the use of reusable packaging, CCEP conducted in-depth analyzes across its operations to better monitor and report progress (Coca-Cola Europacific Partners, 2023b). One such initiative took place in France, where CCEP collaborated with Carrefour to implement a deposit system for RGBs in 150 Carrefour City stores across Paris.

Another case takes place in Coca-Cola Europacific Partners Germany (CCEP DE) who expanded its refillable beverage offerings in 2023 as part of its efforts to reduce packaging waste (Coca-Cola Europacific Partners, 2023e). From September 1-liter RGBs for Fanta, Sprite, mezzo mix, ViO BiO Limo, and Fuze Tea were introduced, following the successful launch of similar bottles for Coca-Cola and Coca-Cola Zero Sugar in 2019.

CCEP DE also standardized the design of all 1-liter refillable bottles to streamline production and increase efficiency (Coca-Cola Europacific Partners, 2023e). Over the previous five years, CCEP DE invested €250 million in reusable packaging, including two refillable glass bottling lines, new refillable packs, an expansion of a refillable bottle pool and new reusable crates. An additional €100 million allocated in 2023 for new bottling lines in Lüneburg and Bad Neuenahr, set to begin operation in 2024.

In Germany, CCEP DE achieved a 97% collection rate for NRGBs through the successful implementation of a DRS (The Coca-Cola Company, 2022). Additionally, the development of a 100% plant-based plastic bottle prototype underscores the company's commitment to innovation in sustainable packaging in Germany (The Coca-Cola Company, 2024b).

4 FRAME OF REFERENCE

This chapter presents theoretical reference essential for understanding RGB systems and sustainability practices in the beverage industry. It reviews CCEPs business model, sustainability initiatives, and circular economy practices, with a focus on Sweden. Key topics include closed-loop supply chains, reverse logistics, and the benefits of RGBs. The role of Deposit Return Systems (DRS) and Reverse Vending Machines (RVMs) in enhancing consumer participation. This framework supports the analysis of strategies for sustainable and efficient RGB system.

4.1 Circular Economy

Circular economy is defined in short as reduction, reuse, recycling and recovery (Kirchherr *et al.*, 2023). In March 2020 the European Commission adopted the new circular economy action plan and is one of the main components in Europe's agenda for sustainable growth, the European Green Deal (European Commission, n.d.). The aim is to reduce waste and increase resource use with over 150 measures focusing on design, durability, reuse, and repair rather than just recycling (Johansson, 2021). It also seeks to phase out single-use plastics and introduce product passports, standardization, and sustainability labels to drive a shift toward a circular economy. While the plan supports recycling and reuse, the EU's other policies still promote resource extraction, driven by geopolitical concerns over critical raw materials. This paradox could hinder the full transition to a circular economy, as increased recycling may coexist with rising extraction, leading to continued environmental harm.

In circular economy ecosystems (CEEs), the engagement of both stakeholders and customers is essential for creating sustainable value (Tabas, 2024). These ecosystems require an interconnected approach where various stakeholders, such as suppliers, manufacturers, and policymakers, coordinate to promote circularity and benefit customers. By moving from isolated actions to a collective ecosystem perspective, CEEs enable resource sharing and mutual support, which enhances the effectiveness of circular economy initiatives.

According to a study on the barriers to CE adoption in the EU, cultural factors are identified as the most significant obstacles (Kirchherr *et al.*, 2018). These include limited consumer interest and awareness. Another study also highlights that customer awareness and engagement play a critical role in advancing CEEs (Tabas *et al.*, 2024). Engaging customers early allows companies to validate and improve circular products, which helps establish sustainable consumption patterns. However, engagement faces specific challenges, such as the limited influence customers have on the design and production processes. Drivers like increased demand for sustainable options and technological advances support this engagement, while barriers like limited consumer input on product design pose challenges.

One critique highlights the limited attention given to understanding what consumers truly value in circular business models and their actual responses to these models (Hobson & Lynch, 2016). It is further indicated that consumers often lack both awareness of and interest in circular offerings, which can hinder their engagement with these initiatives. In a sustainable product-service system the customer's primary role is to accept or reject a new business model (Corvellec et al., 2022).

To increase consumer participation in circular economy practices, companies need to both incentivize and educate consumers (Hobson and Lynch, 2016). Engagement platforms that provide incentives, such as loyalty rewards, can encourage consumers to actively participate in circular practices rather than passively consume. By fostering roles that allow consumers to engage

directly, companies help reshape consumer behavior toward supporting circular economy goals. Educated consumers are more likely to support sustainable practices, as they recognize the environmental impact of their actions, thereby contributing to the system's effectiveness. Efforts to educate consumers on sustainability not only raise awareness but also influence purchasing behaviors (Johansson and Henriksson, 2020). However, consumer-driven change alone has limited impact on systemic shifts in product design. A combined approach with leveraging both regulation and education provide the most effective path toward a fully circular economy.

4.2 Waste Management

In 2022, the EU consisting of 27 countries generated 83.4 million tonnes of total packaging waste, with Sweden contributing 1.36 million tonnes, about 1.63% of the EU's total packaging waste (Eurostat, 2024). For glass packaging waste specifically, the EU produced 15.7 million tonnes, of which Sweden accounted for 254,000 tonnes, representing 1.62% of the EU's glass packaging waste.

The waste hierarchy is a guiding principle in European waste management policy, emphasizing the prioritization of waste treatment methods to minimize environmental impacts (European Parliament and of the Council of The European Union, 2018). Established by the Waste Framework Directive (2008/98/EC), it ranks waste management practices into five levels, shown in Figure 1:



Figure 1. Waste Hierarchy (European Parliament and of the Council of The European Union, 2018)

Prevention: The top priority focuses on reducing waste generation at its source, aiming to avoid creating waste in the first place through sustainable design and consumption.

Reuse and Preparation for Reuse: Involves giving products a second life before they become waste, such as refurbishing or repurposing.

Recycling: This stage includes reprocessing waste materials into new products, maintaining the material's value within the economy.

Recovery: Refers to waste-to-energy processes, like incineration with energy recovery, that extract usable energy from waste.

Disposal: The last resort includes landfilling or incineration without energy recovery, representing the least preferred option due to its environmental impact.

This hierarchy aims to support a circular economy, where materials are kept in use for as long as possible, promoting resource efficiency and minimizing environmental degradation

4.3 Closed-Loop Supply Chain

The combination of Forward Supply Chain (FSC) and Reverse Supply Chain (RSC) is a Closed Loop Supply Chain (CLSC) (Ahmed *et al.*, 2023). The definition of a CLSC is the systems designed to maximize value across a product's lifecycle through dynamic recovery from various returns over time. The CLSC model significantly reduces costs and emissions (Ahmed *et al.*, 2023). In a FSC, materials move from suppliers to manufacturers, then to distribution centers, and finally to customers (Alegoz *et al.*, 2020).

RSC is also known as Reverse Logistics and is defined as planning and management of materials and goods flow from consumption back to origin, aimed at value recovery or proper disposal (Govindan and Soleimani, 2017). In RSC the value is added by the value recovery of returned products (Borumand *et al.*, 2024). One of the challenges in a CLSC is the collection of return products and distribution of end products. A closed-loop supply chain has two primary roles: meeting customer demand through value-added processes and collecting end-of-life products, or returned items, from customers to manage them effectively (Govindan and Soleimani, 2017).

Figure 2 illustrates both forward and reverse supply chain processes. It shows the consumer as a pivotal point in the CLSC, marking the transition between forward and reverse flows (Govindan and Soleimani, 2017)

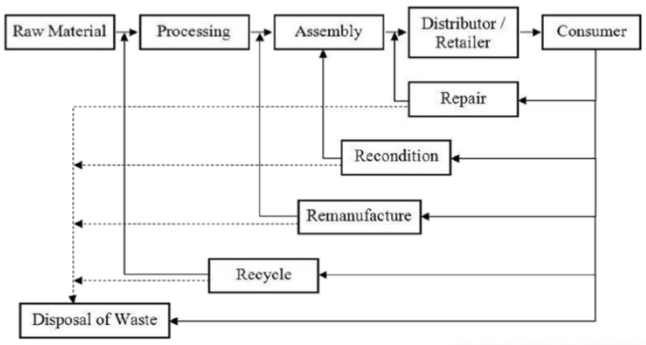


Figure 2. Closed-Loop Supply Chain (Govindan and Soleimani, 2017)

This reverse flow initiated by the consumer in returning end-of-life items enables processes like repair, refurbishment, and recycling (Govindan and Soleimani, 2017). In a CLSC, engaging consumers effectively in the return process is crucial for maximizing product lifecycle value and enhancing sustainable practices, as it allows products to be collected and redirected into environmentally responsible handling stages. The return of used products is the key distinction between open-loop and closed-loop reverse supply chain (Ene and Öztürk, 2014). Used products are typically returned to the original manufacturer in a closed-loop reverse supply chain. In contrast, an open-loop reverse supply chain directs used products to external firms for recovery processes.

Incorporating closed-loop recycling (CLR) within packaging systems offers a dual benefit: it minimizes waste generation by diverting materials from disposal pathways, while also reducing the dependency of industries on raw material inputs (Krzywda, 2022). This cyclical approach to resource management not only supports environmental sustainability but also aligns with industry trends focused on the cyclic use of materials, significantly lowering the environmental impact of packaging at its end-of-life stage. Figure 3 outlines the CLSC network for RGBs (Borumand *et al.*, 2024).

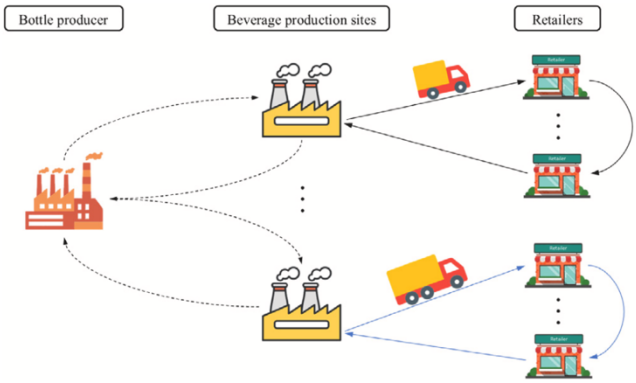


Figure 3. Closed-Loop Supply Chain of Glass Bottles (Borumand et al.*, 2024)*

Production sites manage bottle manufacturing, filling, and sanitation, while retailers handle the demand for filled bottles and the return of empty bottles (Borumand *et al.*, 2024). Distinct vehicle routes link these locations to transport filled and empty bottles. Decisions within this CLSC span three areas: production (acquiring and sanitizing bottles), inventory management (tracking filled and empty bottles), and transportation (vehicle routing and load management).

4.4 Glass Bottles

Globally, RGBs make up 8.8% of TCCC's packaging mix, amounting to 25 billion packages, while NRGBs constitute 1.6%, representing 4.5 billion packages as of 2023 (The Coca-Cola Company, 2023). Within Europe, 85.4% of CCEP's glass packaging is refillable, and 11.6% of PET packaging is in a refillable format (Coca-Cola Europacific Partners, 2024). RGBs represent 10.8% (3,015 million units) of the total packaging introduced into the European market, whereas NRGBs account for 3.4% (960 million units) of a total of 2,558 million units.

Glass is ideal for preserving flavor and maintaining the freshness of products due to its non-reactive, pure composition (Vinci, 2019). Glass bottle usage has increased by 50% in the past decade with a projected rise of 5% by 2026 (Borumand, 2024). Made from natural ingredients, glass is infinitely recyclable, reducing raw material and energy demands with each reuse (Vinci, 2019). Its durability, versatility in design, and high recyclability enhance both brand recognition and environmental sustainability, allowing it to stand out as a closed-loop, eco-friendly packaging solution. While traditionally challenged by weight and fragility, recent innovations have enabled thinner, lighter bottles. These advances not only reduce transport costs but also support sustainability by allowing higher load capacities and lower melting temperatures.

4.5 Refillable Glass Bottles

Life cycle assessments (LCA) have shown that refillable bottles offer a more sustainable and economically beneficial alternative to single-use packaging (Marampoutis *et al.*, 2022). Refillable systems significantly reduce waste and environmental impact compared to single-use options. Another LCA study compared refillable and lightweight containers to single-use glass bottles for wines and spirits in Toronto, Canada (Cleary, 2013). Findings shows that RGBs and aseptic cartons generate the lowest environmental impacts, reducing endpoint impacts by up to 87% compared to conventional single-use glass bottles. The endpoint impacts are divided into ecosystem diversity, human health and resource availability. Furthermore, the largest contributors

to the endpoint level impacts are fossil fuel use and production, transportation and glass production, where fossil fuel use and GHG emissions were between 66%-99% of the three endpoint level impacts. On a municipal scale, replacing conventional packaging with lightweight and refillable options could cut waste by nearly half.

Implementing refillable packaging systems requires modifications across both upstream and downstream processes (Nessi *et al.*, 2015). These adjustments include changes to the supply chain, the adaptation of packaging production methods, and the reconfiguration of transport logistics to efficiently manage returnable and reusable packaging systems. However, the success of refillable bottle systems depends on efficient logistics, particularly in optimizing collection points, capacity, and adapting to fluctuating demand (Marampoutis *et al.*, 2022). Effective management of reverse logistics is vital to ensure the cost-effectiveness and sustainability of refillable bottle systems.

Another analysis of 32 LCAs, where 9 specifically for bottles, concluded that the comparison between RGB and NRGB presented the most significant decrease in CO_2 emissions from total 11 type of packaging analyzed (Megale Coelho, 2020). NRGB has the highest impact compared to PET, aluminum and beverage carton which is mainly due to the production phase. There is an importance of a DRS to collect and refill glass bottles due to 40% of emission associated with glass production are reduced after two or three times of reuse. Furthermore, the overall CO_2 emissions of RGBs were reduced by over a third when reused at least 5 times compared to NRGBs, increasing the number of cycles decrease environmental impact. The bread even points of environmental impact for RGBs compared to glass, aluminum and PET carton is after 2-3rd use.

4.5.1 Deposit Return System

DRS play a crucial role in managing RGBs (Marampoutis *et al.*, 2022). Consumers pay a deposit when purchasing beverages, which is refunded upon returning the empty bottle. This system encourages reuse and helps minimize waste. Figure 4 shows the logistics network for reusable bottles, which involves several stages and stakeholders (Marampoutis *et al.*, 2022).

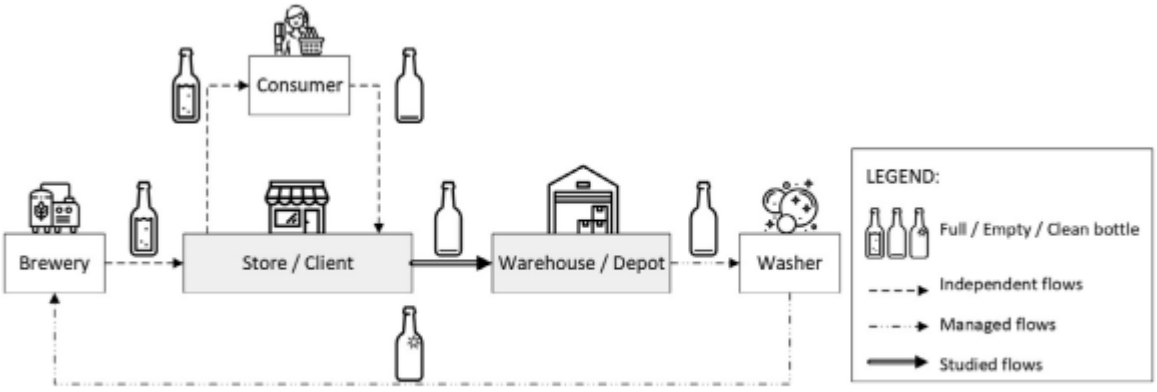


Figure 4. Simplified Logistic Network for RGBs (Marampoutis et al., 2022)

The process starts when the bottle is transported to the brewery for filling and then sent to retailers (Marampoutis *et al.*, 2022). Consumers purchase the bottle with a deposit, encouraging them to return it empty to receive their deposit back. The returned bottle is then collected from the store, transferred to a warehouse, and finally sanitized at a washing station. Once clean, it is sent back to the brewery to restart the cycle.

While DRS improves high quality glass packaging collection, it does not significantly reduce overall glass consumption per capita compared to other management and collection systems

(Agnusdei, 2022). On the other hand, the implementation presents unique advantages that enhance both environmental sustainability in terms of reduction of carbon emissions and economic efficiency (Görgün, 2021). Furthermore, DRS facilitates the clean collection of glass containers, which is crucial for maintaining the high quality of recycled glass materials.

DRS increases return rates by offering financial incentives to consumers (Basuhi, 2024). The effectiveness of DRS for managing glass packaging waste varies by model (Agnusdei, 2022). Agnusdei further points out for a circular business model to succeed, it is essential to have the full cooperation of stakeholders and implement a combination of management strategies and systems tailored to specific contextual factors, such as differences in national cultures. In 2021 the return rate in Sweden were 85% for PET and Metal and in Germany it is 98% for plastic, metal and glass (Görgün, 2021). Three main DRS models are identified, each with unique advantages and challenges (Agnusdei, 2022):

1. Operator-Closing Model: A centralized operator manages the entire process, collecting glass from retailers and overseeing recycling. This ensures efficient handling but does not reduce glass use.

2. Retailer-Closing Model: Retailers handle collection and recycling, simplifying logistics but placing more responsibility on them.

3. Consumer-Closing Model: Consumers return glass directly to collection points, which can boost recycling rates but does not impact overall consumption.

The most effective performance was observed in Iceland exclusively implemented a consumer-closing model with the lowest positive rate in tonnes and a negative rate in kilograms per capita (Agnusdei, 2022). This was followed by the retailer-closing model utilized in Germany and by countries that opted not to implement DRS models but other recycling/recovery programs.

Economically, the DRS for glass containers can lead to substantial cost savings (Görgün *et al.*, 2021). By encouraging the return of glass containers, the system reduces the amount of waste that ends up in landfills, thereby lowering municipal waste management costs and enhancing the material value of recycled glass.

Another critical aspect is the reduction of waste imports (Görgün *et al.*, 2021). Currently, Turkey imports a considerable volume of recycled materials, including glass, to meet its industrial needs. The introduction of a DRS is projected to decrease the import rate of used raw materials fostering local recycling capabilities and contributing to the economic sustainability of the country. This decrease in import reliance not only preserves economic resources but also minimizes the environmental impact associated with transporting materials.

A study conducted in Sweden evaluated two potential DRS for plastic packaging, specifically focusing on food-grade PET trays (Lu *et al.*, 2022). It compared the cost-effectiveness and environmental impact of a recycling-focused DRS and a reuse-focused DRS over a 25-year period. The reuse system provided a more favorable benefit-cost ratio, indicating higher overall economic and sustainability advantages. The reuse scenario shown in Figure 5, shows the HoReCa (Hotel, Restaurant, Cafés) sector, demonstrated significant cost savings in packaging and reduced environmental impacts more effectively than the recycling system.

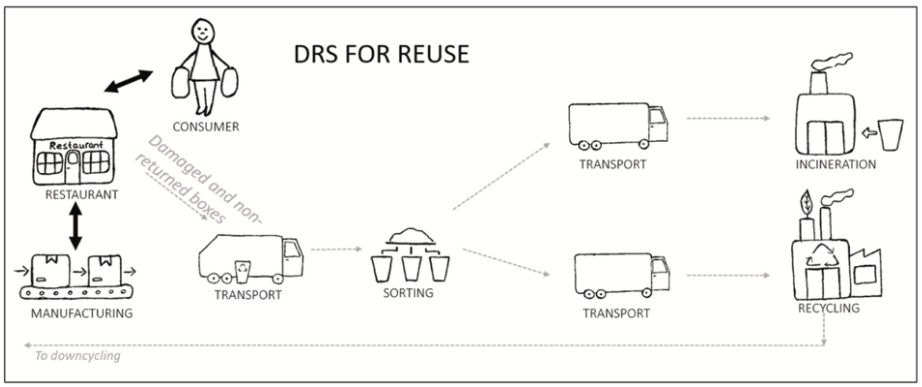


Figure 5. DRS for reuse (Lu et al.*, 2022)*

DRS is aligned with the principles of Extended Producer Responsibility (EPR) (Görgün *et al.*, 2021). Under this policy framework, producers are held accountable for the entire lifecycle of their products, including post-consumer disposal and recycling. This responsibility is crucial for promoting sustainable practices within the beverage industry.

A DRS incorporates two systems for retrieving glass containers (Görgün, 2021). These include manual collection by retailers and the use of Reverse Vending Machines (RVMs). A RVM is a machine where people get a deposit or refund amount for returning empty beverage containers for recycling. It is a key part of container DRS, promoting sustainability (Kokoulin and Kiryanov, 2019). RVMs for PET and metal compact containers whereas RVMs for glass, the containers are processed without breaking it. Manual collection means that the beverage container is accepted by retail staff from consumers (Görgün *et al.*, 2021).

RVMs utilize multiple recognition methods, including material analysis, shape recognition, and barcode scanning, to accurately sort containers (Kokoulin and Kiryanov, 2019). Additionally, RVMs classify containers into categories, including PET bottles, aluminum cans, and glass. This multi-faceted approach is essential for preventing fraud and ensuring proper recycling. Furthermore, effective waste management is increasingly important due to the rising volume of waste and limited landfill space (Tomari, 2016). Recycling serves as a crucial strategy, particularly in areas with restricted disposal options.

4.5.2 Reverse Vending Machines

Current manual recycling practices often deter participation because consumers must separate and transport recyclables to collection centers (Tomari, 2016). Tomari, et al., propose an automated recycling system based on Reverse Vending Machines (RVMs) integrated into standard recycling bins. This system simplifies the recycling process and incorporates a rewards mechanism, making recycling more accessible. The proposed RVM system utilizes sensors to identify user information and weigh the waste, converting the weight into reward points automatically. Implementing a rewards system is critical for encouraging participation in recycling activities, as it has been shown to significantly enhance community involvement. Initial tests of the prototype indicate a 97% success rate in user interactions, suggesting that the RVM framework can significantly enhance recycling participation and efficiency in urban settings.

In another case a RVM incorporates RFID technology to enhance user interaction and transaction efficiency (Dumpayan, 2017). In this system, users are identified through RFID cards, which allow for personalized transactions. When a user approaches the machine with their RFID card, the RVM reads the card to verify the user's identity, enabling them to deposit plastic bottles and accumulate

reward points based on their contributions. Furthermore, another study integrates a reward system to incentivize recycling behavior and pointing out a user-friendly interface to ensure positive experience and encouraging participation (Kim, 2021).

4.6 Innovation in Packaging

Innovative tracking technology is leading the way in sustainable packaging design, contributing to a lower carbon footprint, as highlighted by CCEP (Coca-Cola Europacific Partners, 2021). Advancements in identification and sensor technologies have enabled methods for tracking, tracing, and monitoring packaging conditions (Schilthuizen, 1999). These innovations have paved the way for cost-effective integration of identification systems into packaging, especially for managing RGBs. Radio Frequency Identification (RFID) technology plays a critical role in tracking and tracing, offering both unidirectional and bidirectional options for logistical monitoring. This technology enhances reverse logistics by facilitating the efficient tracking of bottles throughout the supply chain, thus supporting sustainability in refillable packaging systems.

A study on RFID performance with individually tagged wine bottles highlights that tag design and placement significantly influence readability, especially when tags are designed for glass surfaces (Expósito and Cuiñas, 2015). The orientation of bottles relative to RFID readers also affects detection rates, making these factors crucial for optimizing reliable bottle tracking. Another study on RFID tags designed to work on bottles filled with water, addressed issues like interference from the liquid (Rodrigues, 2022). It found that tag performance varies significantly, with reading distances up to 184 cm on empty bottles, reducing to 30 cm when filled. These results highlight the need for customized tag designs for effective traceability in scenarios involving liquid-filled bottles, relevant for inventory management and sustainable returnable bottle systems. Additionally, integrating sensors into packaging allows for monitoring conditions such as temperature and humidity, ensuring product quality and shelf life (Schilthuizen, 1999). However, challenges remain in the standardization and cost-effectiveness of RFID systems, particularly regarding integration into packaging materials and production processes.

4.7 The History of Glass Bottles in Sweden

RGBs were introduced in Sweden in 1886 through a deposit-return system that encouraged reuse and reduced waste (Johansson, 2022). For much of the 20th century, this system dominated beverage packaging, supported by efficient logistics for collection and refilling

However, by the mid-20th century, single-use packaging like cans and PET bottles gained popularity due to convenience, leading to a decline in refillable bottles (Johansson, 2022). In 1994, a law established producer responsibility for collecting and recycling glass packaging. It requires manufacturers and importers to recycle glass sustainably (Svensk Glasåtervinning, 2024). Svensk Glasåtervinning was tasked with setting up a nationwide collection system. Although these new formats were cost-effective, they increased environmental issues, particularly littering (Johansson, 2022). By the late 1990s, refillable bottles were phased out, driven by higher profits from single-use packaging and market liberalization.

4.7.1 Recycling

Swedish glass recycling plays a crucial role in the circular economy, helping conserve resources and reduce environmental impact (Svensk Glasåtervinning, 2024). Managed by Svensk Glasåtervinning (SGÅ) in Hammar, the process involves collecting, processing, and repurposing

around 900 tonnes of glass daily into new raw materials. Proper sorting by consumers, distinguishing between clear and colored glass, enhances efficiency, while contaminants like ceramics and metals are removed using advanced technologies during processing.

Recycled glass reduces energy consumption by about 20% compared to using virgin materials, resulting in lower CO_2 emissions (Svensk Glasåtervinning, 2024). It significantly reduces landfill waste, supporting environmental sustainability (Holmgren and Henning, 2004). Due to its low energy content, recycling is more efficient than incineration for glass, especially when heat recovery systems aren't available. Recycled glass is often blended with new materials to produce new products, promoting a circular economy. Approximately 60% of recycled glass is made into new bottles, while the rest is used in construction materials like glass wool and foam glass (Svensk Glasåtervinning, 2024). As glass can be recycled indefinitely without losing quality, it embodies circular economy principles, reducing reliance on imported raw materials and minimizing transport emissions. This approach enables Sweden to achieve high recycling rates and foster long-term environmental sustainability.

Household recycling behavior in Sweden is influenced by both infrastructure and social norms. Convenient access to recycling facilities, such as property-close collection points, significantly boosts recycling participation by reducing time and effort barriers (Hage, 2009). SGÅ coordinates glass recycling in Sweden, working with municipalities, private contractors, and other stakeholders (Svensk Glasåtervinning, 2024). Municipalities handle household glass waste collection, separating colored and clear glass to maintain quality. From 2027, they will offer property-adjacent collection services to improve access.

Although there is a separation process, it is essential that consumers return the glass correctly (Svensk Glasåtervinning, 2024). Only glass should be inserted, as any waste or foreign material mixed in can hinder the recycling process and compromise the quality of the new raw material. Proper color separation is also crucial; if a colored glass bottle is mixed with clear glass, it affects the production of clear glass, requiring the use of new raw materials to maintain quality. Additionally, if bottles are returned dirty, it can negatively impact the working environment.

In 2022 there was an acute shortage of glass bottles, and consumers were urged to help by not discarding RGBs in regular glass recycling bins (Sveriges Bryggerier, 2022). The demand for glass bottles has been increasing for a long time, but supply has been affected by the COVID-19 pandemic and the war in Ukraine, as both Russia and Ukraine are major producers of glass. This shortage is exacerbated by the mistaken disposal of RGBs, which end up in glass recycling instead of being cleaned and reused. Around 24% of RGBs are still incorrectly discarded each year, particularly 50 cl bottles purchased at Systembolaget, where up to 86% are not returned. Pointing out that in the return crates it is easy to recognize a RGB but when unpacked it’s difficult. During peak periods, such as summer, beverage sales rise by about 30%, making it even more critical for consumers to return bottles so they can be cleaned and refilled in time for increased demand.

Businesses can choose to collaborate with municipalities, deliver glass waste to designated sites, or use private recycling services (Svensk Glasåtervinning, 2024). While SGÅ is not a producer organization, it processes collected glass, refining it into high-quality recycled material. This ensures efficient recycling and supports a circular economy.

Additionally, moral responsibility plays a key role, as individuals who feel a duty to recycle are more likely to participate, even with limited economic incentives (Hage, 2009). While financial measures like volume-based waste fees have limited impact, the study suggests that combining improved infrastructure with informational campaigns can effectively enhance recycling efforts by reinforcing social norms and environmental awareness. Households are also more likely to

recycle when they perceive that their neighbors are doing the same, creating a positive feedback loop in community recycling behaviors.

4.7.2 Deposit Return System for Metal Cans and PET Bottles

The deposit system for metal cans and PET bottles in Sweden, managed by Pantamera, complements glass recycling (Pantamera, 2024). Pantamera driven by the producer responsibility organization Returpack AB and owned by The Swedish Brewers Association (Sveriges Bryggerier), The Swedish Food Retailors Federation (Svensk Dagligvaruhandel) and Livsmedelshandlarna. The ordinance 2022:1274 about producer responsibility for packaging govern and regulate, where the permit and supervisory authority is The Swedish Environmental Protection Agency (Naturvårdsverket).

This system ensures that returned containers are recycled into new, food-grade packaging, significantly reducing carbon emissions. For example, in 2017, 150,000 tonnes of CO_2 emissions were saved through this recycling. Consumers pay a deposit when purchasing these containers, which is refunded upon their return, providing an incentive for recycling. The deposit amounts are SEK 1 for cans and small PET bottles, and SEK 2 for larger PET bottles.

Returpack and Pantamera are two interconnected yet distinct brands, seen in Figure 6 (Pantamera, 2024b).



Figure 6. Returpack and Pantamera Logo

Returpack manages and develops Sweden's recycling deposit system, targeting professional stakeholders with a focus on sustainability, innovation, and operational excellence (Pantamera, 2024b). Pantamera, engages directly with consumers as an encouraging and communicative brand that promotes recycling with a positive, approachable tone. Pantamera acts as the motivator, making recycling accessible and rewarding for individuals, while Returpack is the sustainability expert. The Pantamera logo is used when addressing consumers, and the Returpack logo is used when communicating with business partners, customers, and subcontractors.

Beverage sellers are responsible for ensuring that packaging is correctly labeled with a symbol, seen in Figure 7. The labeling on cans and bottles must indicate that they are eligible for recycling and specify the refund amount, where a small bottle is 1 SEK and a big bottle is 2 SEK (Naturvårdsverket, 2024).

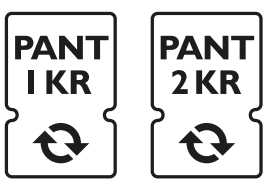


Figure 7. Metal Cans and PET Bottle Return Symbol (Naturvårdsverket, 2024)

In Sweden, grocery stores and cafés/restaurants participate in the deposit system with differing methods. Grocery stores typically install Reverse Vending Machines (RVMs) to collect cans and bottles, receiving a handling fee alongside the refunded deposit (Pantamera, 2024c). They register with the system and maintain ownership of the RVMs. In contrast, cafés and restaurants collect deposit-bearing items manually in bags, which are picked up by suppliers during deliveries (Pantamera, 2024d). They receive compensation after the items are processed at recycling facilities. This approach is more flexible for smaller businesses that do not require RVMs.

From 2023, more packaging including fruit syrup and juice packaging is included in the deposit system (Pantamera, 2024). However, Pantamera's role is limited to handling metal cans and PET bottles, while glass recycling is managed separately through designated collection points. The return rate of cans is 89,5% and PET is 86,2%, and in total 88,5%, with a goal of 90% with a total volume of about 3 million with 2,7 million collected in the DRS system (Returpack, 2023). In total sold bottles, cans account for 70,4% and PET 29,6%.

4.7.3 Deposit Return System for Glass Bottles

Sweden operates a return system for glass bottles, managed by Sveriges Bryggerier (Sveriges Bryggerier, 2024). This system is designed to facilitate the reuse and recycling of glass bottles and is divided into two primary categories: 33 cl and 50 cl glass bottles. The system is administered by Svenska Returglas 33 cl AB Svenska Returglas 50 cl AB, with separate entities overseeing the different bottle sizes. Beverage companies can join this system by entering into service agreements, which grant them access to these standardized bottles and creates made for the system, streamlining the return and reuse process.

Consumers can return empty glass bottles and crates at designated return points throughout Sweden (Sveriges Bryggerier, 2024). Systembolaget and various grocery stores participate in the system, accepting RGBs that display a standardized return symbol, as seen in Figure 8 (Systembolaget, 2024).



Figure 8. Glass Bottle Return Symbol (Sveriges Bryggerier, 2024)

Companies participating in the return system can choose between two types of service agreements (Sveriges Bryggerier, 2024). The Light Service Agreement is designed for companies with lower production volumes, reducing the complexity and administrative burden of participation. It offers a simpler and more flexible structure, allowing smaller producers to contribute to the DRS without the need for extensive management of bottle returns. The Regular Service Agreement is intended for companies with higher production volumes. It requires more commitment and stricter management of bottles and crates.

The system's fee structure depends on the bottle size and the type of service agreement (Sveriges Bryggerier, 2024). For 33 cl bottles, the deposit fee is 0.50 SEK per bottle and 20 SEK per crate when transferred to a third party. For 50 cl bottles, the deposit fee increases to 0.80 SEK per bottle and 25 SEK per crate transferred to a third party. In both cases, there is a crate handling fee of 3 SEK per crate when transferring empty crates.

Administrative fees are determined by the crate balance assigned to each company. These fees are typically lower in the Light Service Agreement due to the reduced administrative responsibilities. In contrast, companies operating under the Regular Service Agreement are subject to more extensive fees, reflecting the higher volume of bottles handled and the need for greater oversight within the system.

4.8 Challenges and Opportunities in a Reuse System

A study outlines key challenges in implementing a national RGB system (Cinderby and McKendree, 2024). One of the primary hurdles is the substantial investment required to adapt or build facilities for processes like cleaning, recoating, and relabeling bottles. Furthermore, market-related challenges such as the low cost of virgin materials and the high initial investments required, further deter businesses from adopting circular practices (Kirchherr *et al.*, 2018).

Additionally, efficient collection and sorting infrastructure is essential for managing the return and recycling processes, ensuring that bottles can be reused effectively (Basuhi, 2024). A major challenge in a RGB network is efficiently collecting bottles from stores and transporting them to the warehouse (Marampoutis *et al.*, 2022). For a viable business model, the price of each cleaned bottle should cover costs like collection, transport, storage, and washing.

A tension exists between the need for standardized bottle designs, which are critical for operational efficiency, and the preferences of consumers and brands for custom designs that reflect brand identity and quality (Cinderby and McKendree, 2024). This conflict complicates efforts to streamline the refillable bottle system. Additionally, incorporating customer feedback early in product development and return processes enables companies to better meet consumer expectations, creating products that are both functional and sustainable (Hobson and Lynch, 2016). Such early involvement ensures that products align with consumer needs.

Logistics further challenges a RGB system, particularly in managing bottle collection and transport. Localized schemes can help mitigate these issues by reducing transport distances, thereby improving efficiency (Cinderby and McKendree, 2024) Collaborations with retailers can also provide convenient return points, simplifying the return process for consumers and encouraging participation (Basuhi *et al.*, 2024).

The environmental benefits of refilling over recycling depend on factors like transport and bottle durability, making the sustainability comparison complex (Cinderby and McKendree, 2024). Effective implementation would also require coordinated policy efforts and regulatory support. While voluntary industry actions contribute to sustainability, effective state-led legislation remains crucial, as it sets enforceable standards that drive consistent participation across companies (Johansson and Henriksson, 2020).

Public participation poses a challenge to the RGB system, as consumers may be reluctant to return bottles, particularly if reused bottles appear less appealing (Cinderby and McKendree, 2024). Addressing this requires targeted awareness campaigns to educate consumers on the environmental benefits of returning bottles, thereby increasing participation rate (Basuhi *et al.*, 2024). There is a growing demand for products that prioritize social responsibility, safety and health (Tabas *et al.*, 2024).

5 SECONDARY DATA RESULTS AND ANALYSIS

This chapter examines the regulatory, environmental, and operational aspects of RGBs and NRGBs under the framework of the EU Packaging and Packaging Waste Regulation (PPWR). By synthesizing insights from legislative directives, lifecycle assessments, and cross-country comparisons, this analysis highlights key challenges, opportunities, and strategies for promoting sustainability in beverage packaging.

5.1 EU Packaging and Packaging Waste Regulation

The regulation on Producer Responsibility for Packaging (2022:1274) plays a significant role in influencing the use of RGBs (Klimat- och näringslivsdepartementet, 2022). This regulation places the responsibility on producers to ensure the sustainability of their packaging, emphasizing recycling and reuse to minimize environmental impact.

A recent EU legislative framework for packaging and packaging waste regulation (PPWR) that are provisionally agreed with the European Parliament and Council are designed to address sustainability and waste reduction across various packaging formats (Council of the European Union, 2024). To encourage the circular economy and sustainable packaging practices, it is important to promote reusable packaging and re-use systems.

PPWR establishes targets for promoting reusable packaging to drive sustainability in beverage packaging (Council of the European Union, 2024). By January 1, 2030, final distributors of alcoholic and non-alcoholic beverages within the EU should ensure that at least 10% of their products are available in reusable packaging systems. Targets for transport packaging, for example pallet wrapping should aim for 40% in a reusable format for the economic operator (Europaparlamentet, 2024).

By January 1, 2040, the reusable packaging target increases to 40%, while transport packaging targets rise to 70% in a reusable format for economic operators (Council of the European Union, 2024; Europaparlamentet, 2024). These targets aim to promote circular economy principles by reducing waste and maximizing material reuse for sustainability.

Member States are permitted to set additional requirements to further the Regulation's objectives, such as improving the purity of collected packaging waste, reducing litter, or promoting circular economy initiatives (Council of the European Union, 2024). In areas with high levels of cross-border commerce, the DRS should facilitate the collection of packaging from other Member States' systems and ensure that deposits charged at the point of purchase can be refunded when the packaging is returned.

If it is technically and economically viable, Member States should strive to ensure that deposit and return systems designed for single-use packaging formats are equally accessible for reusable packaging (Council of the European Union, 2024). They should also evaluate the potential implementation of deposit and return systems for reusable packaging.

5.1.1 Closed- Loop Reuse System

The PPWR emphasizes the need for efficient systems to support RGBs as part of a broader reusable packaging strategy (Council of the European Union, 2024). In these closed-loop systems, packaging remains under the same ownership throughout its lifecycle and is managed by a system

operator and system participants. The system should ensure the packaging is collected, reconditioned, and redistributed efficiently within the system. It should include reverse logistics processes to facilitate the return of packaging from users or end-users to the relevant system participants. The participants of the system should accept returned packaging from designated collection points approved by the system operator, provided it has been used, collected, and stored in compliance with the system's established rules.

To maximize the effectiveness of closed-loop systems, it is crucial to optimize logistics, production, and customer return mechanisms (Council of the European Union, 2024). Addressing bottlenecks in these processes enhances operational efficiency and sustainability outcomes. The regulation encourages best practices to ensure smooth operation across the supply chain.

5.1.2 System Operators

Member States should establish or license a system operator or coordinate activities if multiple operators exist (Council of the European Union, 2024). The system operator, defined as any individual or legal entity is responsible for managing a reuse system within a Member State, oversees its compliance with EU sustainability goals and standards, and should maintain comprehensive documentation (Council of the European Union, 2024). This includes an internal organizational statute, proof of funding mechanisms and regulatory compliance statements. A portion of their annual revenue should be allocated to public awareness campaigns focused on packaging waste management. To ensure effective oversight, operators should supply necessary information to national authorities to facilitate compliance monitoring.

The PPWR promotes DRS to enhance the collection and recycling of packaging, focusing on single-use plastic beverage bottles and metal containers, with encouragement to extend these systems to other packaging types (Council of the European Union, 2024). System operators have the responsibility to operate and establish a DRS and should be an independent non-profit legal entity. DRS incentivizes consumers to return bottles for recycling or reuse, ensuring high-quality recycling and collection. This system complements the broader objectives of the PPWR, enhancing the circularity of packaging materials and contributing to reduced waste.
The DRS should set a minimum deposit level adequate to meet targeted collection rates and outline financial capacity requirements for operators to ensure they can fulfill their responsibilities effectively (Council of the European Union, 2024).

In closed-loop systems, system operators are tasked with maintaining robust tracking and reporting mechanisms (Council of the European Union, 2024). These operators should track the number of rotations each reusable bottle completes, including detailed records of refills, collection rates, and return rates. If precise data is not available, the regulation allows the use of average estimations to meet compliance requirements.

5.1.3 Economic Operators

Economic operator refers to entities such as manufacturers, packaging suppliers, importers, distributors, authorized representatives, final distributors, and providers of fulfillment services (Council of the European Union, 2024). The system operators should provide equal access and fair conditions for all economic operators offering packaging types included in the system.
The reconditioning process for reusable packaging is critical to ensuring that packaging is safe and suitable for reuse. According to Annex VI in the PPWR, economic operators are responsible for implementing a comprehensive reconditioning process (Council of the European Union, 2024). This process involves several key steps: assessing the condition of the packaging, removing any

damaged or non-reusable components, thoroughly cleaning and washing the bottles in accordance with stringent hygiene standards, and making necessary repairs to ensure compliance with both environmental and safety regulations.

To further support sustainability, the regulation stipulates that all reusable packaging should complete a minimum number of rotations to maximize its environmental benefits (Council of the European Union, 2024). Economic operators are responsible for ensuring that their systems are designed to meet these rotation requirements, which are vital for reducing the overall carbon footprint associated with packaging. By ensuring that each bottle is reused multiple times, the environmental impact of the packaging is minimized.

A central element of the PPWR's framework is the fair distribution of costs among participants in the reuse system (Council of the European Union, 2024). The regulation requires that the financial responsibilities related to cleaning, reconditioning, and redistributing reusable packaging are shared equitably among all system participants. This ensures that the cost of maintaining reusable systems is distributed fairly, promoting both economic and environmental sustainability.

Furthermore, sorting capabilities are essential in circularity of packaging where technological innovations should be encouraged, where core objective of the PPWR is to promote packaging innovation with the aim of reducing packaging waste (Council of the European Union, 2024). The PPWR grants packaging with innovative, environmentally beneficial features an additional five years to meet recyclability requirements. Justification and recycling plans should be provided in technical documentation, and economic operators should notify the European Commission and relevant authorities before market release.

5.1.4 Design

To align with the sustainability objectives set by the PPWR, packaging should be designed to reduce environmental impact across their entire lifecycle (Council of the European Union, 2024). This includes ensuring that bottles are reusable multiple times and recyclable at a high quality at the end of life.

The design of glass bottles should facilitate the highest possible number of rotations while maintaining safety, quality, and hygiene requirements during the processes of being emptied, unloaded, refilled, or reloaded while also having minimum trips or rotation, standardized designs and requirements for re-use (Council of the European Union, 2024).

All packaging included in a reuse-system should be clear and easily recognizable labeled to ensure that end users can easily understand the packaging as returnable and understand the need to return it (Council of the European Union, 2024). Furthermore, ensuring that consumers are aware it should be returned through designated collection channels approved by national authorities, and any associated fees should be transparently disclosed. The packaging can also feature a harmonized color label.

The PPWR mandates strict regulation of harmful substances, such as lead, cadmium, and mercury (Council of the European Union, 2024). This ensures that the packaging remains safe throughout its lifecycle, minimizing environmental and health risks. The regulation also supports minimizing packaging materials to avoid unnecessary waste to ensure packaging circularity.

Furthermore, manufacturers are required to meet specific targets for recycled content in glass packaging by incorporating recycled materials into their products (Council of the European Union, 2024).

5.1.5 Consumer and End-Users

In PPWR there is a section about information on managing and preventing packaging waste where end-users are an essential (Council of the European Union, 2024). End users should be able to return packaging at the original point of sale or nearby. Final distributors should accept reusable packaging of the same type, form, and size as they market, free of charge, within the designated reuse. They are responsible for facilitating recovery and return throughout the distribution chain. Furthermore, final distributors should redeem deposits according to the reuse system rules.

To effectively implement PPWR, consumers should be protected from misleading or unclear information regarding packaging attributes and appropriate disposal methods (Council of the European Union, 2024). Ensuring packaging within the extended producer responsibility scheme by a harmonized symbol accessible across the scheme's territory through a QR code or standardized digital marking. This symbol should clearly inform consumers about the reuse and recyclability of the packaging and indicate the producer's compliance with extended producer responsibility obligations. If the DRS is mandatory, the packaging should bear a label informing customers about collection channels.

Producers and producer responsibility organizations should actively educate consumers on preventing and managing packaging waste (Council of the European Union, 2024). This includes providing information on re-use options, label explanations, and proper disposal methods.

Consumers should also be informed that packaging labeled as compostable is only suitable for industrial composting at bio-waste facilities, not for home composting. Emphasis should be placed on responsible waste management, highlighting the importance of avoiding littering. Modern technology should be leveraged to communicate this information, utilizing traditional methods like posters and social media campaigns, as well as innovative approaches such as QR codes on packaging to provide detailed electronic access to information (Council of the European Union, 2024).

5.2 Sweden's Beverage Industry RISE LCA Study

A LCA study conducted by RISE (Research Institutes of Sweden) investigated the climate impact of different packaging systems (Löfgren et al., 2024). The study specifically measured CO_2 emissions per liter of beverage, providing a comprehensive analysis across the entire life cycle of each packaging type. Leading beverage industry players, including Spendrups, Carlsberg Sweden, Coca-Cola Europacific Partners Sweden AB, and Åbro, participated in the study. The assessment encompassed all stages of the packaging life cycle: from raw material extraction and production to collection, recycling, and reuse of materials, or disposal as waste. For the RGB system, additional processes such as bottle washing for reuse were included, whereas for the single-use NRGB system, glass recycling was a key factor.

The climate impact was calculated using a 100-year time horizon (GWP100) in line with the Intergovernmental Panel on Climate Change (IPCC) characterization method for Global Warming Potential (Löfgren *et al.*, 2024). The results of this analysis are further detailed in the corresponding analysis section.

The distribution of glass bottles in the Swedish market for 2022 of the participating companies, shown in Table 3, presents notable changes since 2018 (Löfgren *et al.*, 2024).

Table 3. Glass Bottles Distributed in the Swedish Market 2022 (Löfgren et al., 2024)

Bottle Type	Units Distributed	Change (2018–2022) (%)
33cl RGB	34 074 260	-41,5%
50cl RGB	38 668 395	-3,8%
33cl NRGB	169 783 398	+7,9%
50cl NRGB	39 509 096	+3851%

The total glass bottle market for the involved companies in the LCA study grew from 256.8 million units in 2018 to 282 million units in 2022, an increase of 9,8% (Löfgren *et al.*, 2024). However, this growth was largely driven by single-use glass bottles. The distribution of 33 cl RGBs declined by 41,5%, and 50 cl RGBs decreased by 3,8%. In contrast, 33 cl NRGBs increased by 7,9%, while 50 cl NRGBs surged by 3851%. Overall, the total number of RGBs fell by 26,1%, while NRGBs grew by 32.2%.

5.2.1 Carbon Emissions

The study revealed that NRGBs have the highest climate impact in beverage distribution, primarily due to their energy-intensive production phase (Löfgren et al., 2024). Emissions were analyzed across the entire life cycle, including primary packaging production, brewery processes, transportation, secondary packaging, and collection. As shown in Figure 9, the 33 cl RGB demonstrated the lowest emissions among the glass options, while the 150 cl PET exhibited the lowest impact overall.

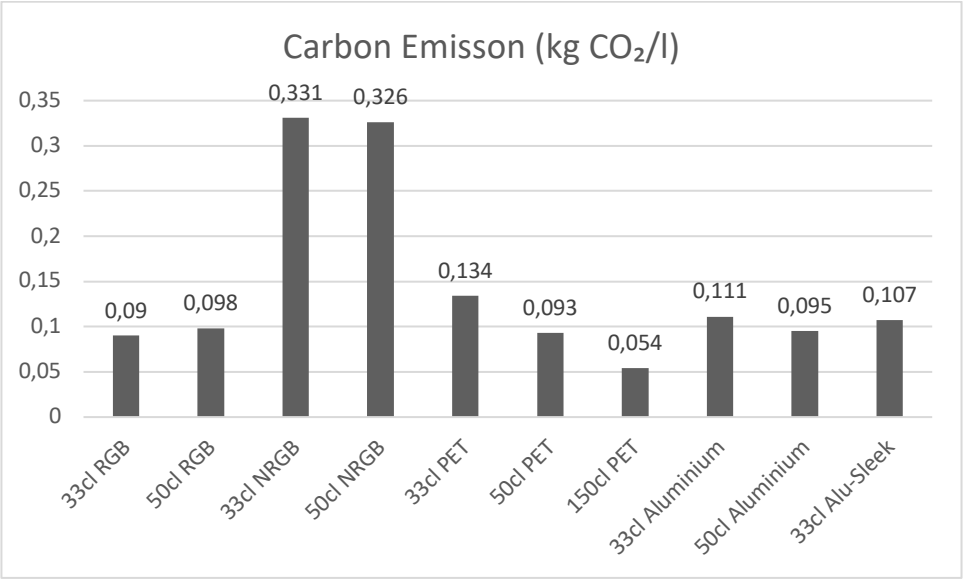


Figure 9. Carbon Emission per Packaging Type (Löfgren et al.*, 2024)*

In comparison, 33 cl PET showed higher emissions than all RGB types, and the 50 cl PET outperformed only the 50 cl RGB but still emitted more than the 33 cl RGB (Löfgren et al., 2024). Across all packaging systems, NRGBs consistently displayed the highest CO_2 emissions, highlighting their environmental inefficiency. The lower emissions of RGBs are attributed to their ability to undergo multiple reuse cycles before recycling, significantly reducing the need for new material production. As seen in Table 4, NRGBs emitted approximately 3.7 times more CO_2 per

liter for 33 cl bottles and 3.3 times more for 50 cl bottles compared to their reusable counterparts. For 33 cl bottles, transitioning from NRGBs to RGBs reduces emissions per bottle with a 73% decrease and for 50 cl bottles the reduction with a 70% decrease.

Table 4. Glass Bottle CO_2 Emissions (Löfgren et al., 2024)

Bottle Type	Carbon Emisson (kg CO_2/l)	Change (Year 2018-2022)
33cl RGB	0,090	-4,4%
50cl RGB	0,098	-8,5%
33cl NRGB	0,331	-1,8%
50cl NRGB	0,326	-3,5%

Between 2018 and 2022, emissions for RGB systems were influenced by transportation, distribution, and cleaning processes (Löfgren et al., 2024). However, RGB emissions decreased during this period, with 33 cl RGBs achieving reductions through material efficiency and improved secondary packaging, and 50 cl RGBs benefitting from lighter caps, reduced secondary packaging, and extended lifespans. For NRGBs, emissions reductions were primarily driven by production efficiency and increased use of recycled materials. The 33 cl NRGB achieved improvements through lighter materials and optimized secondary packaging, while the 50 cl NRGB saw reductions due to lighter caps and refined secondary packaging. Additionally, collection-related emissions for NRGBs declined as SGÅ adopted lower-impact fuels.

Using the baseline values for total volume and units distributed from the RISE LCA, additional calculations were performed to model the Swedish market based on the involving companies (Löfgren *et al.*, 2024). RGB and NRGBs were analyzed into their packaging type being reusable or single use. RGBs accounted for 26% of the total units distributed and 29% of the total volume, while NRGBs dominated with 74% of the units and 71% of the total volume, shown in Table 5.

Table 5. Swedish Market 2022 Units and Volume Distributed

Bottle Type	Units Distributed	Units Distributed (%)	Volume (l)	Volume (%)
RGB	72 742 655	26	30 578 703	29
NRGB	209 292 494	74	75 783 069	71
Total	282 035 149	100	106 361 773	100

In 2022, glass bottles, including NRGBs and RGBs, accounted for a total volume of 106 million liter, with the 33cl format, RGBs represented only 17% of the total volume, with NRGBs dominating at 83%, as shown in Table 6. This underscores the dominance of single-use bottles in this format. In contrast, the 50cl format showed higher adoption of RGBs, which accounted for 50% of the total volume.

Table 6. Swedish Market 2022 Volume

Bottle Type	Total Volume (l)	Volume RGB (l)	Volume RGB (%)
Glass Bottle 33 cl	67 273 027	11 244 506	17
Glass Bottle 50 cl	39 088 746	19 334 198	50

The data highlights a significant opportunity to increase the adoption of RGBs, especially in the 33 cl format, where their share of total units and volume remains relatively low. The difference in volume compared to units is largely attributed to the higher proportion of larger bottle formats within RGBs, which can deliver more volume per unit compared to smaller bottles that dominate NRGBs. Smaller bottle formats require a higher number of bottles to achieve the same total volume

compared to larger formats. As a result, RGBs optimize beverage delivery with fewer bottles, reducing logistical demands such as transport frequency and associated emissions.

From a carbon emission perspective, the current system of mixed RGB and NRGB resulted in total emissions of 27,9 million kg CO_2 in 2022, as seen in Table 7.

Table 7. Swedish Market 2022 CO2 Reduction 100% NRGB to RGB

Bottle Type	Carbon Emisson RGB (kg)	Carbon Emisson RGB (%)	Carbon Emisson NRGB (kg)	Current Carbon Emisson (kg)	New Carbon Emisson All RGB (kg)
Glass Bottle 33 cl	1 012 006	5	18 545 441	19 557 446	6 054 572
Glass Bottle 50 cl	1 894 751	23	6 439 983	8 334 734	3 830 697
Total	**2 906 757**	**10**	**24 985 423**	**27 892 180**	**9 885 270**

If all NRGBs transitioned to RGBs, emissions could be reduced to 9.9 million kg CO_2, based on the carbon emission intensity from the LCA (Löfgren *et al.*, 2024). When further examining the current RGB share of emissions and volume, RGBs account for 29% of total volume but only 10% of total carbon emissions, highlighting their efficiency in reducing carbon footprints compared to NRGBs.

Breaking it down by bottle size, 33 cl bottles represent 17% of total market volume but account for 5% of carbon emissions. This demonstrates their strong environmental potential and positions them as a key format for expanding RGB adoption. Meanwhile, 50 cl bottles, which held 50% of total volume, contribute only 23% of total carbon emissions. This reflects the advantages of economies of scale in the RGB process and presents a valuable opportunity for further expansion.

With further calculation based on bottle type the findings highlights the substantial environmental benefits of transitioning from NRGBs to RGBs, with a total carbon emission reduction of 18 million kg CO_2, representing a 65% decrease, shown in Table 8. When broken down by bottle size, 33 cl bottles demonstrate the greatest reduction, with a 69% decrease in emissions amounting to 13.5 million kg CO_2. Similarly, 50 cl bottles achieve a 54% reduction, equivalent to 4.5 million kg CO_2.

Table 8. Swedish Market 2022 CO2 Reduction NRGB to RGB

Bottle Type	Reduction Carbon Emission (kg)	Reduction Carbon Emission (%)
Glass Bottle 33 cl	13 502 874	69
Glass Bottle 50 cl	4 504 037	54
Total	**18 006 911**	**65**

5.2.2 Transport and Energy

The analysis showed a positive trend toward increasing the use of energy sources with reduced climate impact across all stages of the beverage systems (Löfgren et al., 2024). For NRGBs, the climate impact of collection was reduced due to the use of lower-emission fuels by collection services, emphasizing the role of sustainable fuel choices. In the conversion and filling processes, a higher share of "green electricity" was utilized. Notably, all electricity used in breweries in 2018 and 2022 was renewable, with the 2022 mix achieving a lower climate impact due to an increased share of hydropower. The study also noted that emissions from the cleaning and refilling stages at

breweries increased due to changes in the energy mix used during these processes, though this rise had a limited effect on the overall lifecycle impact. This emphasizes the importance of the type of energy used in cleaning operations.

During the distribution, collection, and recycling stages, a greater proportion of renewable fuels such as HVO (hydrotreated vegetable oil) was used, replacing diesel to lower the climate impact (Löfgren *et al.*, 2024). However, specific data on HVO usage remained unavailable, and the analysis assumed a distribution distance of 250 km from brewery to consumer, consistent with the 2018 study. Beverage weight was included in transport calculations, but beverage production was excluded. A sensitivity analysis assessed four transport scenarios to evaluate the climate impact of 100% diesel versus 100% HVO fuel usage. The results, summarized in Table 9, demonstrated that HVO consistently led to lower emissions across all scenarios.

Table 9. Emission of Fuel Type Scenario (Löfgren et al., 2024)

Scenario	Description	Case Diesel (CO_2 kg/11,000 l)	Case HVO (CO_2 kg/11,000 l)
1	**Base Scenario**: Full delivery to a single customer, returning empty.	839	229
2	**Equal Volumes to All Customers**: Three stops with an empty return.	645	177
3	**Smaller Initial Volumes**: Three stops with smaller deliveries at the start.	677	185
4	**Larger Initial Volumes**: Three stops with larger deliveries at the start.	500	108

Among the scenarios, Scenario 4 with HVO, larger deliveries made at the start of the route, achieved the lowest climate impact at 108 kg CO_2/11,000 liters (Löfgren *et al.*, 2024). Scenario 3 with HVO followed as the next most efficient option, emphasizing the benefits of using renewable fuels like HVO for distribution, significantly reducing emissions compared to diesel. Additionally, the improved renewable electricity mix in 2022, driven by a higher share of hydropower, further contributed to lowering the climate impact of brewery operations.

5.2.3 Packaging Design

The study also discusses the role of material use and transportation in the environmental impact of glass bottles (Löfgren *et al.*, 2024). RGBs are heavier than NRGBs to withstand multiple use cycles, which increases transport-related emissions. However, the overall savings from the reuse of these bottles outweigh the increased transport emissions. Furthermore, the report provides data on weight reductions shown in Table 10. The weight of the 33cl RGBs decreased by about 0.2%, while the 50cl refillable bottles saw a 0.1% reduction compared to previous studies (Löfgren *et al.*, 2024).

Table 10. Bottle Type Weight Reduction (Löfgren et al., 2024)

Bottle Type	2022	Weight Change
33cl RGB	284,5 g	-0,2%
50cl RGB	344,6 g	-0,1%
33cl NRGB	196,8 g	-1,9%
50cl NRGB	299,7 g	-0,1%

These weight optimizations contribute to a slight decrease in transport emissions due to reduced energy consumption during transport. However, despite these weight reductions, refillable bottles remain heavier than single-use bottles due to the durability required for multiple uses.

Additionally, the return crates for 33cl bottles have been optimized, resulting in a 42% weight reduction, while those for 50cl bottles experienced a 48% reduction (Löfgren *et al.*, 2024). These substantial reductions in the weight of the crates help to further decrease the overall transport-related emissions for the RGB system, making the logistics more efficient and sustainable.

The environmental impact analysis for RGBs included the contribution from each trip of the reusable crate, calculated as a weighted material factor (Löfgren *et al.*, 2024). As outlined in Table 11, secondary packaging material, for RGBs decreased by 42% to 48% per bottle over their lifespans. The weight of bottle caps is set at about 2 grams and was also decreased from year 2018 to 2022 but is excluded here.

Table 11. Secondary Packaging Weight Reductions (Löfgren et al., 2024)

Bottle Type	Weight per Bottle (g)	Change (2018–2022) (%)
33cl RGB	9,7	-18,8
50cl RGB	18,7	-6,7
33cl NRGB	0,93	-42
50cl NRGB	1,2	-48

In addition to weight reductions, the plastic crates used for RGB transportation were assigned a lifespan of 40 years in the 2022 analysis (Löfgren et al., 2024). Each crate completed approximately 3.5 trips annually between brewery and customer, representing a 3.4% increase in usage since 2018 for the 33 cl RGB shown in Table 12.

Table 12. Crate Life (Löfgren et al., 2024)

Crate Type	Annual Trips	Annual Trips Change (2018–2022) (%)	Crate Lifespan (years)	Lifespan Change (2018–2022) (%)
33cl RGB	3,5	+3,4%	40	+100%
50cl RGB	40	+75%	40	+100%

This extended lifespan resulted in a 50% emissions reduction despite a 3% increase in crate weight for the 33 cl RGB (Löfgren *et al.*, 2024). Similarly, the 50 cl RGB crates achieved a 9% reduction in secondary packaging weight, further lowering emissions. For NRGBs, the reductions in packaging weight also led to significant emissions decreases. The 33 cl NRGB achieved a 19% weight reduction, resulting in a 24% decrease in emissions, while the 50 cl NRGB saw a 7% weight reduction and a corresponding 48% decrease in emissions.

5.2.4 End-of-Life

Recycling rates for glass in Sweden remain high, with about 86% of glass being recycled in 2022 (Löfgren *et al.*, 2024). Although this high rate reduces the environmental impact of NRGBs, it cannot fully match the emission savings achieved through the reuse of refillable bottles, which require far less frequent production. According to the report, the collection of empty bottles often utilizes the same trucks that deliver filled bottles, improving logistical efficiency.

A certain percentage of refillable bottles are lost during the reuse cycle and are replaced with newly manufactured bottles (Löfgren *et al.*, 2024) . Based on their estimated lifespan, a 33cl RGB carries

the environmental burden of producing 7.1 grams of glass (equivalent to 1/40th of a bottle), while a 50cl RGB carries 43.1 grams (or 1/8th of a bottle). Any refillable bottles not collected through the return system are processed like other glass waste, through public recycling facilities.

The proportion of recycled glass used in bottles ranged between 50% and 90%, with colored glass typically containing a higher proportion of recycled content (Löfgren *et al.*, 2024). For the recycling of secondary packaging, such as the crates used with refillable bottles, it is assumed that these materials follow standard recycling processes for plastic packaging. In 2022, 33% of collected plastic packaging was recycled, while the remaining material was directed to incineration.

5.3 Deposit Return System for Refillable Glass Bottles

By addressing infrastructure gaps and leveraging store chains, Sweden could continue to optimize the RVM network, improve return rates, and advance its recycling goals. Sveriges Bryggerier provides valuable insights into the geographical and organizational spread of RVM units, which plays a significant role in understanding the infrastructure available for the DRS (Sveriges Bryggerier, 2024). With their data an analysis of the store chain and distribution of RVM units, seen in Table 13.

Table 13. Store Chains and Units (Sveriges Bryggerier, 2024).

Store Chain	RVMs (unit)
ICA	471
COOP	435
Willys	163
Hemköp	94
City Gross	39
Other	35
Total	**1237**

As of 2024, there are 1 237 RVMs in Sweden, managed by Tomra (Sveriges Bryggerier, 2024). These machines are integral to Sweden's DRS, ensuring the collection and recycling of RGBs. Out of these 436 RVMs accept both RGBs and crates and 801 RVMs accept only RGBs. The largest chain, ICA, accounts for the highest number of RVM units, with 471 machines in operation, followed by COOP with 435 units. These two chains cover over 73% of the total RVM infrastructure, indicating their dominant role in facilitating consumer participation in the recycling process. Willys and Hemköp have significantly fewer units, with 163 and 94 RVMs respectively, while City Gross operates 39 units. Additionally, other smaller chains and independent stores account for the remaining 35 RVMs.

The data emphasizes the critical role of store chains like ICA and COOP, which host the largest number of RVMs, in driving the success of the DRS. These stores serve as pivotal touchpoints for engaging customers, enhancing the consumer experience, and increasing participation. They present a significant opportunity to further improve return rates through targeted initiatives.

5.3.1 Delivered Volume

An analysis of the cities with the highest NRGB volumes provides critical insights into consumer demand, logistical challenges, and recycling infrastructure efficiency. This assessment focuses

exclusively on the Home Segment (retailers), excluding the HoReCa segment, to emphasize consumer-driven demand and the associated DRS challenges.

Locations within greater metropolitan areas, such as Åkersberga, Sollentuna, and Täby, rank prominently in delivered volumes (Internal Coca-Cola Europacific Partners, 2024). Similarly, regional and remote locations, including Skellefteå and Luleå, also exhibit notable delivery volumes. This highlights the importance of a robust infrastructure in suburban and regional areas to ensure nationwide recycling efficiency and avoid bottlenecks in the system.

The average transportation distance for 33 cl NRGB deliveries to the cities is approximately 356 kilometers, driven by deliveries to geographically dispersed and remote locations. This underscores logistical challenges, particularly the need to balance infrastructure expansion with efforts to reduce carbon emissions associated with transportation.

5.3.2 Population and Density

An analysis of delivered NRGB volumes and the distribution of RVMs across Sweden's ten largest cities highlights significant disparities in recycling infrastructure accessibility and efficiency. Table 14 provides a detailed comparison of population, RVM density (units per 100,000 inhabitants) and utilization rates (bottles per RVM), offering valuable insights into infrastructure performance relative to population needs.

Table 14. Home Segment NRGB 0.33 l by Delivered Volume Largest Cities Sweden

City	Population	Density (RVMs/100 000 population)	RVMs (units)	Utilization Rate (bottles/RVM)
Stockholm	988 943	2,3	23	642
Göteborg	604 616	5,0	30	173
Malmö	362 133	6,9	25	192
Uppsala	245 329	7,3	18	105
Linköping	167 404	7,2	12	148
Västerås	159 662	8,1	13	111
Örebro	159 348	10,0	16	158
Helsingborg	151 306	5,3	8	198
Jönköping	146 161	8,9	13	133
Norrköping	145 163	4,8	7	158

Stockholm, the city with the highest delivered volume, faces significant strain on its infrastructure. With a low RVM density of 2.3 units per 100,000 inhabitants and a high utilization rate of 642 bottles per RVM, highlights for a possible network expansion to accommodate its nearly one million residents effectively.

In contrast, Göteborg and Malmö demonstrate better-balanced systems. Göteborg, with the highest number of RVMs (30), achieves a density of 5,0 units per 100,000 inhabitants and a low utilization rate of 173 bottles per RVM. Similarly, Malmö's density of 6,9 and utilization rate of 192 reflect efficient infrastructure management. These examples suggest that cities with smaller populations can still maintain effective recycling systems through proportionate infrastructure planning.

Smaller cities such as Örebro, Västerås, and Jönköping further exemplify efficient systems, with RVM densities between 8.1 and 10.0 and proportionate utilization rates. However, cities like Helsingborg and Norrköping indicate potential areas for improvement, as their lower RVM densities may limit accessibility.

Interestingly, Uppsala, with 18 RVMs and a density of 7,3, surpasses Linköping despite their similar populations. This suggests more targeted efforts in Uppsala to enhance consumer participation in the DRS. Additionally Skellefteå and Luleå underscore the need for robust infrastructure in remote areas to address logistical challenges and reduce carbon emissions from transportation. The average distance from CCEP Jordbro to the listed cities is approximately 273 kilometers, further emphasizing the need to optimize the placement of RVMs to improve accessibility while minimizing transportation emissions.

The analysis reveals a potential misalignment between delivered NRGB volumes and RVM availability in suburban areas, such as Åkersberga, Sollentuna, and Täby, which rank prominently in delivered volumes could indicate a potential unproportionate RVM coverage. Addressing these gaps through infrastructure expansion and strategic placement of RVMs in suburban and under-served areas is critical for improving recycling accessibility and participation rates.

6 INTERVIEWS RESULTS AND ANALYSIS

This section summarizes the key findings from interviews, categorized into themes and subthemes, to provide an analysis of insights related to RGB systems. The themes reflect stakeholder perspectives on investments, behavior, supply chain dynamics, sustainability, and innovation, highlighting opportunities and challenges in implementing efficient and sustainable RGB systems, shown in Table 15.

Table 15. Interview Themes and Subthemes

Theme	Subthemes
Investments	Sustainable Innovation, Supply Chain
Behavior and Engagement	Knowledge Gap, Convenience and Accessibility, Incentives and Communication
Supply Chain Management	Logistics, Infrastructure, Production
Refillable Glass Bottles	Deposit Return System, Packaging Efficiency, Market Demand
Sustainability	Supply Chain Practices, Collaboration, Strategic Alignment, Circular Economy Principles
Key Stakeholders	Industry Responsibility, Collaboration, Knowledge Sharing
Innovation	Technological Advancements, Technology for Sustainability

6.1 Investments

"An innovation that is just sustainable is not sustainable in itself if we're going to lose money and go out of business"

The transition to RGB systems necessitates significant investments, encompassing financial, logistical, and operational challenges which emerged from the interviews. A key theme was the necessity for substantial modifications to production lines. These changes include installing or upgrading machinery for sorting and cleaning bottles, as well as adapting or replacing existing infrastructure to accommodate RGB systems. Stakeholders underscored the complexity of these adjustments, emphasizing the need for detailed planning and significant financial resources to ensure seamless integration.

Transport logistics emerged as another critical challenge in the transition to RGBs according to the interviewees. The design of RGB pallets, while durable for repeated use, results in fewer bottles per pallet and, consequently, higher per-unit transport costs. Stakeholders noted that specialized logistics systems, including robust pallets and tailored transport networks, add further to the overall expense. Additionally, stakeholders recommended double-stacking pallets during return trips. Since empty pallets are lighter, stacking them two high effectively doubles transport volume per journey, significantly reducing costs and improving space utilization. Stakeholders also proposed exploring transport cost offsets, such as subsidies or cost-sharing mechanisms, to alleviate the financial pressures associated with RGB logistics. These strategies collectively contribute to enhancing the economic and environmental sustainability of the system.

To address these challenges, several strategies were proposed from the interviews. One approach was load consolidation, which involves grouping smaller deliveries to maximize transport capacity and reduce the frequency of trips. This strategy is particularly advantageous for smaller customers, as it optimizes logistical efficiency and reduces the environmental footprint of transport operations. Route optimization was also highlighted as a key measure, with streamlined routes minimizing transit times, reducing fuel consumption, and lowering logistics costs. Collaborative

logistics—where companies share transport resources—was identified as an innovative method to enhance efficiency across the industry.

Stakeholders emphasized the critical importance of balancing economic and environmental goals to ensure the long-term viability of RGB systems. While environmentally focused initiatives are essential, their financial feasibility should not be overlooked, as economically unsustainable models risk jeopardizing business continuity. Comprehensive strategic planning was repeatedly identified as crucial, ensuring that initial investments align with long-term gains and that both sustainability and financial metrics are achieved.

The analysis highlighted the need for innovations that address the dual objectives of environmental and economic performance. Ensuring compatibility with existing operational frameworks is vital to minimizing disruptions and avoiding inefficiencies during the transition.

6.2 Behavior and Engagement

"We've noticed that many customers aren't even aware that they can return glass bottles, and for those who do, they don't understand why it's better than recycling. There's a significant knowledge gap we need to address."

From the interviews, stakeholders identified three essential drivers for the successful adoption of RGB systems: convenience, incentives, and communication. Accessible return locations were emphasized as vital for participation. One interview highlighted Returpack's efforts to ensure recycling opportunities even in remote areas, such as archipelagos and mountain resorts, as a strength.

It was noted by interviews that the RGB system for the HoReCa sector is well functioning but noted a larger gap within the end-user and consumer segment. Interviews revealed that the current deposit value for RGB is considered insufficient to motivate consumers. Stakeholders recommended increasing deposit levels or introducing creative rewards to strengthen consumer motivation. Insights from pilot programs highlighted the effectiveness of such approaches; for instance, a lottery wheel incentive successfully boosted return rates for reusable cups. Conversely, pilots with unclear messaging or insufficient rewards faced low engagement, underscoring the importance of aligning program design with consumer expectations.

Enhancing messaging about return processes also emerged as a top priority from the interviews. Stakeholders noted that many consumers find existing communication for RGB unclear, leading to confusion and reduced participation in the reuse system. Further pointing out the success of "Panta Mera" campaign, which uses relatable and positive narratives to normalize recycling, was frequently cited as an example of the power of effective messaging.

Stakeholders from the interviews emphasized that many consumers are unaware of the existence of RGB systems and the environmental benefits of reusing bottles over recycling, which impacts return rates. They pointed out that younger consumers, particularly men, were highlighted as being less familiar with how the system works. This underscores the need for targeted educational campaigns and tailored outreach to address these knowledge gaps and improve participation. Stakeholders also recommended analyzing low-engagement customer groups to design initiatives that address specific barriers.

Mentioned in the interviews, Sweden faces distinct cultural and operational barriers to the widespread adoption of RGB systems. Interviewees from countries such as Germany pointed out

that decades of familiarity have embedded RGB return behaviors into daily routines, where other stakeholders further noted that Swedish consumers lack these ingrained habits. Stakeholders noted that fostering long-term behavioral change in Sweden requires integrating sustainable practices into everyday life through consistent education and engagement.

Expanding RGB systems requires coordinated efforts among key stakeholders, including retailers and distributors. Insights from interviews emphasized the importance of proactive partnerships to enhance consumer participation. Interviewees highlighted that collaboration with retailers, particularly through tailored campaigns aligned with their sustainability goals, is an effective strategy for increasing engagement. Furthermore, staff engagement was also repeatedly noted in interviews as a key factor. Well-trained retail staff who actively promote and explain the benefits of RGB systems could influence consumer behavior to ensure consistent and effective communication with consumers.

The sustainability benefits of RGB systems rely heavily on achieving high return rates. Stakeholders noted that the return is important to justify the environmental impact of the system, where experiences from international pilots in Chile revealed challenges with bottles completing only three cycles before being removed from the reuse system. Furthermore, insights from interviews revealed that pilot projects in a smaller Swedish town have demonstrated the potential for improving consumer engagement through visible in-store communication, informational campaigns, and physical displays. Interviewees identified low return rates as a persistent issue and emphasized the need to test solutions in strategically chosen locations. Trials conducted in less central areas highlighted the importance of carefully selecting pilot project sites to maximize participation and gather meaningful insights. They further stressed that aligning incentives and communication strategies with consumer behavior is crucial to achieving long-term success.

6.3 Supply Chain

"Since we only have one production site in Jordbro, but our packages are all the way up to the north of Sweden, that would mean we also have to get them back."

Logistical challenges present significant obstacles to the implementation of RGB systems, as highlighted in interviews. Stakeholders noted that Sweden's infrastructure for managing RGB systems is comparatively smaller and less developed than in countries with well-established systems. With a single production site in Jordbro and bottles distributed nationwide, the process of returning bottles from remote areas adds complexity.

Interviewees stressed the critical need for substantial investments in handling, washing, and storage facilities to address these gaps and ensure compliance with regulatory requirements. They further emphasized that integrating RGB systems alongside existing NRGB production lines requires meticulous planning to maintain operational efficiency. Balancing the distinct needs of both systems is essential to prevent disruptions and enable seamless production transitions.

Transportation to northern regions, where customer density is lower, requires tailored solutions to balance environmental sustainability and cost-effectiveness as mentioned by stakeholders. Interviewees suggested concentrating RGB systems efforts around urban centers, such as Stockholm, to streamline operations and reduce inefficiencies. Stakeholders also emphasized the importance of integrating two-way transport routes and using advanced tools to optimize regional performance, minimizing costs and CO_2 emissions. Furthermore, the usage of third-party logistics providers, such as DHL, has shifted operational dynamics for CCEP and its competitors as

mentioned in interviews. While it reduces internal fleet management burdens, it underscores the importance of collaboration to ensure efficient bottle collection and redistribution.

Interviewees highlighted that glass's weight and bulk add significant complexity to storage and transportation compared to cans or PET, requiring specialized infrastructure and substantial storage capacity. Unlike compactable materials, glass demands considerable space and resources, making its logistics more costly and operationally intensive. Stakeholders noted that current safety protocols are sufficient for transporting dirty bottles alongside other goods, eliminating the need for additional measures.

Stakeholders compared RGB logistics to the Returpack system, which efficiently collects volumes through compacting machines installed in over 3,200 grocery stores and an additional 5% via "Panta Mera Express" stations as mentioned in an interview. The centralized compactor vehicle model was highlighted for lowering operational costs. However, interviewees noted that RGB systems face challenges due to collection systems. Batch returns, a strategy utilized in the Returpack system, were identified by interview as an effective way to reduce logistical challenges. By scheduling returns, the frequency of smaller, inefficient shipments could be minimized, improving both cost-effectiveness and sustainability.

Interviewees recommended measures to further optimize transport efficiency, such as redesigning pallets to increase bottle capacity per trip and adopting improved crate structures. These changes would address current inefficiencies, reduce costs and CO_2 emissions, and enhance system performance. Additionally, innovations in packaging design, such as improved trays, were emphasized as critical for reducing trips and logistical expenses while advancing sustainability goals.

Efficient sorting processes are essential for RGB systems. Stakeholders noted inefficiencies in sorting practices, emphasizing the need for automation and upgraded machinery. From the interviews, a comparison to Germany revealed challenges in sorting at multiple stages, including manual processes at after the RVM stage, where bottles were often misallocated. Improved sorting systems could enhance operational efficiency, reduce bottlenecks, and ensure bottles are properly prepared for reuse.

Managing seasonal fluctuations was highlighted critical for the efficiency of RGB systems, particularly during high-demand periods like summer from the interviews. The importance was noted to increasing storage capacity and implementing proactive inventory management strategies to prevent supply chain disruptions. Efficient handling of both full and empty bottles is essential to maintaining seamless operations and ensuring system reliability. Restaurants and bars benefit significantly from closed-loop systems, where breweries deliver and collect bottles directly. This streamlined approach simplifies bottle management, ensures higher return rates, and enhances operational efficiency. Stakeholders highlighted the importance of maintaining and expanding such closed-loop systems to support scalability and sustainability. Furthermore, proactive collaboration with customers was identified as a key strategy for managing seasonal demand. Interviewees stressed the need for improved planning and communication to anticipate changes in demand and prevent stock shortages during peak periods. By aligning inventory strategies with customer needs, RGB systems can remain efficient and responsive, even during periods of increased activity.

6.4 Refillable Glass Bottle System

"In 1976, 61% of the total volume was returnable glass, but in 2024, it's under 2%."

Stakeholders highlighted in interviews that RGBs, which accounted for 61% of the market in 1976, have drastically declined to under 2% in 2024. This reduction was attributed to shifts in consumer behavior, infrastructure challenges, and regulatory changes. Interviewees noted that the removal of RGB systems in Sweden was driven by logistical concerns, high water usage, and the need for upgrades to glass production lines. These past challenges underline the importance of addressing logistical, environmental, and operational considerations when planning the reintegration of RGB systems.

According to interviews, RGB bottles are designed for durability across multiple reuse cycles, with 33 cl bottles reusable up to 40 times and 50 cl bottles up to eight times. While these reuse rates demonstrate clear environmental benefits, interviewees highlighted the trade-off posed by thicker materials, which increase carbon output and overall material usage. Stakeholders emphasized the need to balance operational durability with sustainability to ensure successful implementation. In contrast, NRGBs are optimized for single use, making them lighter and easier to transport without requiring the durability of RGBs. Managing dual return systems involving both RGBs and NRGBs were identified as a significant logistical challenge. Interviewees stressed the importance of optimizing transportation, storage, and handling processes to reduce costs and improve operational efficiency.

Current infrastructure and consumer habits further complicate scaling RGB systems. Stakeholders pointed out limited return rates and the lack of perceived brand value by producers as major barriers. To address these issues, interviewees highlighted the need for standardized packaging to simplify production and logistics while allowing for branding flexibility.

Interviewees also emphasized that the long-term sustainability potential of RGB systems depends on strategic investments in modernized infrastructure, logistics optimization, and consumer engagement. Overcoming past obstacles and addressing current challenges are essential to maximizing the environmental and operational benefits of RGB systems.

6.5 Sustainability

"We switched all our transport companies to run on HVO100, cutting almost 90% of our emissions from transportation"

Insights from interviews highlighted CCEP's significant progress in sustainability, particularly in transportation, packaging, and recycling. A key achievement discussed by stakeholders is the company's transition to HVO100 fuel, reducing transport-related CO_2 emissions by nearly 90%. Interviewees emphasized that further optimization strategies, such as maximizing load capacities, implementing efficient routing methods, and monitoring trip counts, are essential for aligning logistics efficiency with sustainability goals. Efforts to improve the sustainability of the RGB system were also a key focus in the interviews. Stakeholders emphasized strategies such as optimizing routes, monitoring trip counts, and maintaining HVO100 fuel use to reduce emissions and enhance lifecycle sustainability. Furthermore, electric vehicles would be further analyzed for implementation in CCEP's logistics network to complement existing strategies by driving emissions even lower and aligning with evolving environmental regulations.

Despite the environmental benefits of RGB systems, interviewees identified challenges in their cleaning processes. The bottle washing required for RGBs consumes substantial water and energy, increasing their environmental impact. Stakeholders pointed to the use of green electricity in Sweden as a partial solution but emphasized the need for innovative water-saving technologies and energy-efficient systems to enhance the sustainability of bottle-washing operations.

Interviewees also highlighted Returpack's engagement in international sustainability initiatives, where it shares best practices with other Nordic countries nations on establishing recycling systems. These collaborations not only enhance global sustainability efforts but also refine local operations, driving continuous improvement.

Circular economy principles emerged as a cornerstone of CCEP and TCCC's sustainability strategies, consistently emphasized by stakeholders throughout this research. Recycling and reuse are pivotal to reducing waste, minimizing environmental impact, and promoting long-term resource efficiency. Stakeholders highlighted the importance of alignment between TCCC and CCEP to ensure cohesive and effective execution of these strategies across all operational levels. Tailored campaigns that resonate with customer sustainability objectives were also emphasized as a critical success factor. Stakeholders stressed that such initiatives not only advance environmental goals but also reinforce partnerships by aligning with shared values. Additionally, CCEP's alignment with the European Union's climate neutrality targets emerged as a key strategic advantage, ensuring the organization remains relevant and competitive within the evolving regulatory landscape.

Innovation was identified from an interview as a seamlessly integrated element within CCEP's broader sustainability and business framework, underscoring its role as a vital component of a holistic strategy rather than a standalone initiative. This integration enables CCEP to align its transport, packaging, and recycling practices with circular economy principles, driving progress toward its ambitious science-based climate goals: halving its climate impact by 2030 and achieving net-zero emissions by 2045.

As mentioned in interviews, CCEP's use of 100% recycled PET in packaging demonstrates a strong commitment to reducing its carbon footprint and supporting a circular economy. Stakeholders highlighted that using recycled PET significantly lowers emissions compared to virgin materials by keeping resources within the system and minimizing waste. This circular approach can also be applied to glass packaging, which the demand for raw materials and energy-intensive production processes, further decreasing carbon emissions while enhancing resource efficiency.

6.6 Ownership

"The industry has been given the responsibility to solve this on its own."

The interviews emphasized a self-regulated approach regarding the DRS for cans and PET bottles in Sweden which has been entrusted to the beverage industry. One interview noted that Returpack's DRS has evolved into one of the most efficient recycling systems globally since its inception in 1984. Originally focused on mitigating the environmental impact of aluminum cans, the system has expanded to include PET bottles and now processes over 2.7 billion packaging units annually. Interviewees noted that Returpack operates as a collaborative, non-profit entity, prioritizing environmental and operational efficiency over profit generation. This model fosters transparency and sustainability while encouraging stakeholders to establish a self-sustaining system that achieves high recycling rates and operational excellence.

Stakeholders emphasized the critical distinction between industry-organized and state-mandated DRS. Returpack's industry-led model grants beverage producers, retailers, and other stakeholders' direct responsibility and control over the system's operations, promoting accountability and efficiency. Interviewees also pointed out external political influences as complicating the implementation and scaling of RGB systems. They emphasized that the success of these systems places significant demands on consumers to engage actively in recycling efforts. To address this, robust public education campaigns and strategies for consumer engagement were deemed crucial for fostering participation and building long-term habits. Proactive engagement with regulatory bodies was highlighted as essential for addressing logistical and compliance challenges. Stakeholders suggested that aligning supply chain processes with regulatory requirements requires close coordination to streamline operations and ensure system efficiency.

Additionally, the standardized financing model ensures financial sustainability by preventing any one party from disproportionately benefiting, according to interviews. By maintaining industry ownership, Returpack and its partners have successfully built a recycling system that aligns with environmental goals while ensuring operational efficiency, which the interviewees highlighted this model as a key factor behind the system's global reputation for effectiveness and innovation.

Interviewees consistently emphasized collaboration as a critical factor in successfully adopting RGB systems. Stakeholders underscored the importance of engaging customers, (retailers, HoReCa and wholesalers) and consumers, through tailored incentives. Expanding RGB systems, stakeholders explained, requires coordinated efforts among stakeholders. Collaboration fosters consistency and helps create a unified, scalable framework that can address systemic inefficiencies. Logistical partnerships were also considered vital for maintaining high return rates. Interviewees noted that streamlining collection and sorting processes is essential for ensuring operational efficiency while aligning business goals with environmental objectives. Stakeholders emphasized building partnerships across the supply chain, with wholesalers and retailers playing pivotal roles in managing bottle return logistics. Tailored incentives and collaborative efforts with municipalities and regulatory bodies were seen as essential for overcoming logistical and compliance challenges.

From the interviews there was a divided view on RGB systems. While some highlighted the efficiency and familiarity of aluminum cans and PET packaging, and the effective NRGB system others saw the sustainability potential of RGB systems. Stakeholders agreed that the success of RGB systems hinges on systemic changes, strong consumer demand, and consistent engagement.

The importance of international collaboration in improving recycling systems was pointed out by interviewees. Sweden has shared its expertise in DRS management while learning from neighboring countries such as Norway and Denmark. Germany's established RGB system was also frequently mentioned as a benchmark. Interviewees emphasized that exchanging expertise with countries experienced in RGB implementation fosters continuous improvement. They noted that drawing lessons from other countries, such as Nordic neighbors and Germany, Sweden can strengthen its recycling infrastructure and position itself as a leader in sustainable recycling practices. This collaborative exchange of knowledge not only enhances Sweden's ability to overcome existing hurdles but also creates opportunities for regional leadership in recycling systems.

6.7 Innovation

"There is potential in using tracking technologies for refillable glass systems, though it requires effort over time"

Centralized data systems, DataMatrix, RFID, QR codes, and digital transformation were repeatedly highlighted in the interviews as tools for improving lifecycle management, sustainability, and consumer engagement. Stakeholders emphasized the potential in tracking technology to improve reverse logistics logistical planning, resource optimization, and consumer engagement, enhancing the efficiency of RGB system management, However, the caution against overreliance were noted, emphasizing these tools as a complement to existing logistical frameworks. Interviewees emphasized that successful implementation requires seamless integration with existing systems, logistical frameworks, and consumer-centric strategies. A balanced implementation strategy that integrates technological advancements with operational efficiencies was noted as critical for achieving long-term sustainability. Other identified key challenges from the interviews were reducing marking times, ensuring scalability, and achieving cost-effectiveness.

Pilot tests insights from the interviews, using RFID chips demonstrated successful real-time tracking of reusable cups, illustrating the applicability of this technology to reuse systems. These pilots showcased the potential of advanced tracking systems to efficiently manage multiple usage cycles while enhancing transparency and lifecycle monitoring. Furthermore, stakeholders pointed out challenges such as inefficiencies in recycling technology and resource-intensive processes. While smaller-scale applications have demonstrated potential, scaling these systems to handle high volumes and complexities at a national or global level requires continuous development, refinement, and substantial investment. Balancing these innovations with practical logistical strategies is critical to ensuring long-term sustainability and operational success.

Automation and digital transformation in logistics workflows also emerged as crucial areas for improvement. Interviewees highlighted increasing the percentage of digital orders to 95% as a key objective for reducing manual processing and improving efficiency. By adopting digital systems, organizations could enhance transparency, streamline workflows, and optimize the management of RGBs.

The technological foundation for advanced tracking systems, including unique coding and scanning mechanisms, is already established as noted from the interview. Centralized data management systems play a pivotal role in monitoring the lifecycle of RGBs, as noted by interviewees, but could be a challenge due to the amount of data. By linking codes to a central database, stakeholders could track bottles across multiple use cycles, improving reverse logistics and enhancing transparency. Tracking the movement of bottles throughout their lifecycle was highlighted as a factor for sustainability. Proper trip monitoring could help identify inefficiencies, reduce bottle loss, and optimize reverse logistics. Despite this potential, stakeholders highlighted that the current RGB infrastructure lacks the necessary scanning mechanisms to fully operationalize these tracking systems. This deficiency affects the ability to monitor returnable bottles effectively and undermines the overall sustainability of the system.

7 DISCUSSION AND CONCLUSION

This chapter presents a discussion of the findings and conclusions drawn during the master's thesis. The analysis intends to address the research questions and provide insights into the potential of RGB systems in Sweden to answer the research questions.

7.1 Discussion

The analysis of interviews, secondary data and literature highlight several critical steps for the successful implementation of RGB systems in Sweden. Addressing consumer knowledge gaps, increasing deposit values, improving communication, and fostering collaboration among stakeholders are essential for building a sustainable and scalable system. By aligning incentives and messaging with consumer expectations and investing in infrastructure improvements, Sweden could maximize the environmental and economic benefits of RGB systems and ensure their long-term viability.

One aspect of this investigation as identified through the analysis, is the balance between regulation-driven innovation and business-driven innovation. Regulation-driven innovation by directives like the PPWR, sets mandatory targets that encourage industries to adopt sustainable practices, such as standardized bottle designs or DRS (Council of the European Union, 2024). These regulations challenge businesses to rethink their strategies and invest in technologies that meet both environmental and legislative demands. Conversely, business-driven innovation could arise from a proactive mindset within companies that seek to differentiate themselves and create value beyond compliance. This type of innovation often focuses on enhancing consumer experience, improving operational efficiency, or strengthening brand identity through unique and sustainable packaging solutions, which were topics from the interviews.

The analysis of the findings reveals that the interplay between regulation-driven and business-driven innovation underscores the importance of fostering an innovation-driven mindset among all actors in the system. Regulations, such as those outlined in the PPWR, establish a foundation and create urgency for action by setting mandatory targets. These regulations could be interpreted as opportunities for businesses to drive innovation rather than as obstacles. A collaborative approach, where governance frameworks align with business ambitions, is essential to ensuring that innovation becomes a shared goal rather than a source of conflict.

7.1.1 Ownership and Industry Collaboration

The interviews emphasized that successful RGB implementation requires collaboration among stakeholders in the industry including producers, retailers, municipalities, governing bodies, and consumers. Tailored incentives, educational campaigns, and regulatory alignment were noted as essential for fostering consumer participation and achieving sustainability goals. External stakeholders highlighted retailer campaigns, staff training, and clear in-store communication as key strategies, where internal stakeholders noted underscored pilot projects and the importance of strategic site selection and clear messaging to maximize return rates. This aligns with CEEs where stakeholder and customer engagement enhance the collective effectiveness of circular economy principles (Tabas, 2024). Furthermore, success in circular business models depends on full stakeholder cooperation and strategies tailored to the specific context (Agnusdei, 2022).

An industry-wide collaboration compelling stakeholders to address the challenges of implementing DRS systems aligns closely with insights from the interviews about Returpack, Returglas, and Tomra, which supply RVMs. As noted in the interviews, RVMs currently handle

primarily PET and cans, with limited capacity for glass. For an RGB system to function effectively, significant scaling and modernization of the DRS infrastructure is essential. This aligns with findings that simplified recycling processes, combined with incentives, have successfully maximized return rates in DRS systems (Tomari, 2016). Incorporating user-friendly technology interfaces could further ensure a positive experience and encourage broader participation (Dumpayan, 2017).

The environmental benefits of refilling over recycling are influenced by factors such as transport efficiency and bottle durability, making the sustainability comparison between systems complex (Cinderby and McKendree, 2024). As highlighted in the interviews, upgrading RVM technology, enhancing storage and sorting capabilities, and optimizing logistics are critical to realizing these benefits. Leveraging Returpack's established brand to include RGBs could build on existing infrastructure and consumer trust, streamlining implementation. Alternatively, a centralized entity modeled after Returpack's success could oversee the RGB system. Such an entity would unify operations, optimize logistics, and foster collaboration among stakeholders, ensuring a cohesive approach aligned with EU sustainability goals.

While voluntary industry actions play a role in advancing sustainability, coordinated policy efforts and regulatory support are essential to drive consistent participation. As Johansson and Henriksson argue, effective state-led legislation sets enforceable standards that compel companies to align with environmental objectives (Johansson and Henriksson, 2020). This is supported by interviews emphasizing the need for robust governance frameworks to define stakeholder roles, standardize financing models, and promote accountability within RGB systems.

As highlighted in the interviews, Sweden's Returpack exemplifies the effectiveness of collaborative, non-profit models in achieving high recycling rates and operational efficiency. Scaling RGB systems within this framework requires clearly defined stakeholder responsibilities, strong partnerships, and regulatory alignment. Interviewees emphasized that ownership structures like Returpack's, which distribute accountability among key stakeholders, are crucial for fostering collaboration and ensuring system success. These models effectively align interests while addressing the logistical and operational challenges of RGB implementation.

While Sweden has a centralized entity in Returglas for RGBs, it has not operated with the same effectiveness as Returpack, referring to lower return rates and smaller volumes. As highlighted in the interviews, establishing a modern centralized entity could offer significant advantages for RGB systems. Such a structure would unify operations, streamline logistics, and enhance collaboration among stakeholders, ensuring alignment with EU sustainability goals. However, the interviews also emphasized that adapting Returpack's ownership model to RGB systems would require addressing specific challenges, including glass durability, the complexities of return logistics, and the need for robust consumer engagement. A well-designed RGB system has the potential to balance sustainability and operational efficiency. Yet, as underscored in the interviews, achieving this balance will require coordinated industry efforts and supportive policy frameworks to overcome these hurdles and secure long-term success.

7.1.2 Investments

Transitioning to RGB systems requires substantial financial, logistical, and operational investments according to interviews. The adaption of facilities of cleaning, recoating and labeling bottles was as a primary challenge (Cinderby and McKendree, 2024). Interviewees emphasized the need for balanced strategies that align sustainability goals with cost-effectiveness. Upgrades to production lines and tailored transport logistics, such as double-stacking pallets and route optimization, were identified as critical but costly measures to enhance efficiency. Where

mentioned in the interviews that the current system is effective, further confirms that the high initial investments and the low cost of virgin material could hinder business from adopting circular models (Kirchherr *et al.*, 2018). Strategic planning and alignment with existing frameworks are vital to minimizing disruptions and ensuring long-term success. By adopting innovative solutions and fostering collaboration, businesses could build a sustainable RGB system that balances economic and environmental objectives. Collaborative logistics and cost-sharing mechanisms were suggested as viable ways to reduce expenses while maintaining environmental benefits. Financial sustainability is essential, as economically unviable models risk failure despite their environmental advantages.

The interviews revealed that while the current system is effective, high initial costs and the low price of virgin materials could be significant barriers to adopting circular models as mentioned by a study (Kirchherr et al., 2018). The finding from the interviews and the PPWR highlighted collaborative logistics and cost-sharing mechanisms as viable strategies to mitigate expenses while preserving environmental benefits. Strategic planning and alignment with existing frameworks, as emphasized in the interviews, are critical to minimizing disruptions and ensuring long-term success. By fostering stakeholder collaboration and implementing innovative solutions, businesses could develop a sustainable RGB system that effectively balances economic viability with environmental objectives.

7.1.3 Sustainability

The interviews and analysis emphasize the significant environmental advantages of RGBs, primarily due to their extended reuse cycles, which offset emissions from transportation, production, and cleaning processes (Löfgren et al., 2024). Stakeholders highlighted that RGB systems consistently produce lower CO_2 emissions compared to NRGBs, PET bottles, and aluminum cans. However, the thicker materials used in RGBs result in higher production emissions, whereas NRGBs, though lighter and easier to transport, generate significantly higher emissions due to their single-use nature. The sustainability of RGB systems hinges on addressing operational challenges, as identified in interviews. Handling losses, quality rejections, and premature bottle damage were noted as significant barriers, increasing per-use emissions when bottles do not reach their full reuse potential. Stakeholders emphasized the importance of improving material durability, implementing robust handling processes, and optimizing logistics systems to mitigate these inefficiencies and maximize the environmental performance of RGB systems.

RGBs were identified in both the interviews and literature as consistently emitting less CO_2 than NRGBs across all sizes (Löfgren et al., 2024). NRGBs require new production for each use, leading to substantially higher emissions, making them less viable as a long-term sustainable solution. For example, 33 cl PET bottles emit more CO_2 than any RGB size, while 33 cl RGBs, with extended lifespans, were noted in the interviews as particularly efficient. In contrast, 50 cl RGBs, which have shorter reuse cycles, contribute to higher emissions. Stakeholders and literature emphasized that extending the lifecycle of larger RGB bottles would significantly improve their sustainability, aligning with principles of maximizing product lifecycle value and minimizing end-of-life impacts (Govindan and Soleimani, 2017; Borumand et al., 2024). Additionally, emissions from glass production decrease by 40% after two or three reuse cycles, with overall emissions dropping by over a third when RGBs are reused at least five times (Megale Coelho, 2020).

Circular economy principles, as frequently emphasized in the interviews, are central to CCEP's strategy, focusing on recycling, reuse, and waste reduction. The success of 100% recycled PET packaging, mentioned by stakeholders, offers a scalable model for glass packaging, promoting resource efficiency and reducing emissions. Implementing an RGB system would align with

CCEP's "This is Forward" strategy and TCCC's "World Without Waste" initiative, which target at least 25% of global beverage volume in refillable or returnable formats (Coca-Cola Europacific Partners, 2023; The Coca-Cola Company, 2023). This approach reflects circular economy goals of reduction, reuse, recycling, and recovery (Kirchherr et al., 2023).

Stakeholders noted that while single-use glass bottles benefit from high recycling rates, reuse-focused systems provide greater economic and sustainability advantages. This finding is supported by studies comparing reuse-focused and recycling-focused DRS systems, where reuse demonstrated superior performance in both areas (Lu et al., 2022). Transitioning to RGBs, as indicated in the interviews, would enable CCEP to enhance its sustainability performance, align with circular economy objectives, and achieve long-term environmental benefits.

7.1.4 Supply Chain

Implementing refillable packaging systems requires modifications across both upstream and downstream processes, these adjustments include changes to the supply chain, the adaptation of packaging production methods, and the reconfiguration of transport logistics to efficiently manage returnable and reusable packaging system (Nessi *et al.*, 2015). This was confirmed from the interviews where they highlighted an establishment of the necessary infrastructure for an efficient DRS, including collection points, storage facilities, and transport networks, requires meticulous planning. Suboptimal routes or inefficient handling processes could undermine the sustainability benefits of RGBs. Efficiently collecting bottles from stores and transporting them to warehouses remains a major challenge in an RGB network (Marampoutis et al., 2022).

Interviews emphasized that collaboration with logistics providers and customers is vital for creating a resilient, sustainable supply chain to support RGB systems. Retailer partnerships for return points were identified as critical to encouraging consumer participation in the reuse system (Cinderby & McKendree, 2024). Additionally, Sweden's vast geographical distances and dispersed population pose significant challenges for transportation and reverse logistics, increasing emissions and logistical complexity. Both the literature and interviews highlighted the need for significant supply chain adjustments, including changes to production, transport, and logistics processes, to implement RGB systems effectively (Megale Coelho, 2020).

The reverse supply chain plays a central role in enabling a circular economy (Krzywda, 2022). The RISE LCA study assumes RGBs are transported using fully loaded 22-ton diesel trucks over an average one-way distance of 250 km. However, the analysis and interviews revealed significant variability in actual transport distances within Sweden, ranging from 25 km (Jordbro to Stockholm) to 475 km (Jordbro to Gothenburg). The average distance for high-volume cities is approximately 264,5 km, which aligns with the baseline. However, this calculation does not account for return transport distances or volume-weighted influences, highlighting the need for region-specific logistics strategies to minimize emissions.

The reliance on a single production site in Jordbro exacerbates logistical challenges, particularly for remote areas. Interviews emphasized the importance of investments in washing, storage, and handling facilities, as well as optimized two-way transport routes, to improve efficiency and reduce costs. Combining forward and reverse supply chains into a closed-loop supply chain (CLSC) is essential for operational efficiency and aligns with regulatory requirements for closed-loop reuse systems, as outlined in the PPWR (Council of the European Union, 2024; Ahmed et al., 2023). Efficient sorting and automation were underscored in interviews, for eliminating bottlenecks and ensuring reuse of bottles. Seasonal demand fluctuations necessitate proactive inventory management and expanded storage capacity to maintain supply chain reliability. Closed-

loop systems in the HoReCa sector offer scalable solutions for improving return rates and streamlining operations.

The RISE study also omitted an analysis of return transport distances, leaving emissions from reverse logistics unclear. Stakeholders highlighted the importance of optimizing logistics networks, including the use of regional hubs to facilitate bottle collection and reduce transport inefficiencies. Maximizing truckload capacity and prioritizing shorter, high-density delivery routes were identified as effective strategies to reduce the carbon footprint (Löfgren et al., 2024).

CCEP has demonstrated leadership in sustainability by transitioning to HVO100 fuel, which reduces CO_2 emissions by up to 90% compared to traditional diesel (Sjöholm, n.d.). Further optimization through route planning, load consolidation, and adoption of electric or hybrid vehicles could enhance these benefits. Stakeholders also suggested exploring rail transport for longer routes and collaborating with other companies to optimize logistics, reduce costs, and collectively lower emissions. By leveraging renewable electricity and increasing hydropower usage, emissions could be further reduced.

The energy mix in Sweden plays a crucial role in minimizing environmental impacts, particularly in bottle cleaning processes. Breweries that utilize renewable or low-carbon energy sources significantly reduce the overall carbon footprint of RGB systems. However, the RISE LCA's allocation of 10% energy consumption to cleaning was critiqued for oversimplifying actual requirements. Stakeholders stressed the need for more precise data to enhance lifecycle assessments, as well as investments in energy-efficient and water-saving technologies to further reduce emissions and improve sustainability.

7.1.5 Lifespan, Reusability and Waste

Recycling and recovery processes were highlighted in interviews as crucial for managing bottles at the end of their lifecycle. Recycling ensures that glass from damaged or unusable bottles remains in circulation, preventing resource depletion. The lifespan of RGBs is central to their sustainability. RISE LCA data indicate that a 33cl RGB has an environmental burden equivalent to 7,1 grams of glass per cycle (1/40th of a bottle), while a 50cl RGB accounts for 43,1 grams per cycle (1/8th of a bottle) (Löfgren et al., 2024). These figures highlight the efficiency of reuse systems, as increasing reuse cycles reduces the carbon footprint per bottle. However, interviews revealed uncertainties regarding the estimated lifespans of 40 cycles for 33cl bottles and 8 cycles for 50cl bottles. Analysis suggests that extending bottle durability and optimizing designs could further enhance environmental performance, an opportunity consistently emphasized by stakeholders.

DRS and RVMs were consistently identified in interviews as essential for managing RGB systems effectively, a finding supported by literature (Marampoutis et al., 2022). PPWR sets ambitious reuse percentage targets that significantly shape DRS design. These targets align with the European waste hierarchy, prioritizing waste prevention as the most effective strategy (European Parliament and Council of the European Union, 2018). While RGB systems extend the lifecycle of bottles and reduce the need for frequent production, interviews clarified that they fall under the "preparation for reuse" category, as they do not eliminate the production phase entirely.

Stakeholders raised the need for clarity regarding PPWR reuse targets, particularly whether these should be based on the volume of beverages sold or the number of units returned. A volume-based approach would likely encourage larger bottles, as fewer units are needed to meet reuse targets. However, interviewees noted that this could reduce consumer convenience and limit single-serving options, potentially influencing consumer behavior and market dynamics. Conversely, a unit-based approach, also highlighted in analysis, would emphasize smaller, portable bottles to meet

reuse criteria. While aligning with consumer preferences, this approach introduces logistical challenges, as smaller bottles require more extensive collection, sorting, and washing efforts, potentially elevating transportation demands and costs.

7.1.6 Design and Brand Identity & Technology Potential

The PPWR recommends adopting standardized bottle designs, presenting both opportunities and challenges for Coca-Cola (Council of the European Union, 2024). The iconic Coca-Cola glass bottle, originally designed in 1915 to be instantly recognizable, remains a cornerstone of the brand's identity (Paulsson, 2024). Glass' high recyclability and durability further enhance brand recognition and sustainability (Vinci, 2019). However, interviews highlighted concerns that standardized designs could dilute Coca-Cola's unique identity, as custom designs are integral to reflecting brand heritage and quality (Cinderby & McKendree, 2024). Stakeholders emphasized incorporating customer feedback early in product development to balance consumer expectations with operational efficiency (Hobson & Lynch, 2016).

Internally, resistance to standardized designs may arise due to their potential impact on Coca-Cola's branding. Nonetheless, interviews suggested that repositioning glass bottles as sustainable packaging could strengthen Coca-Cola's leadership in sustainability. Aligning packaging designs with environmental goals supports compliance with the PPWR while bolstering the brand's image as environmentally responsible. With consumer expectations increasingly prioritizing sustainability, this integration could yield significant environmental and business benefits.

However, standardized designs offer significant advantages, particularly in enhancing operational efficiency within DRS (Cinderby and McKendree, 2024). Interviews emphasized that standardization simplifies logistics and production processes, addressing major barriers to scaling RGB systems. Investments in automated washing facilities and advanced recognition technologies, such as barcode scanning and material analysis, were identified as critical to achieving efficiency at scale (Kokoulin & Kiryanov, 2019). Shifting Coca-Cola's branding focus to labels rather than bottle shapes could allow unique designs to coexist with standardized systems while enhancing operational efficiency.

Logistical challenges, such as the weight and bulk of glass bottles, require innovative solutions to improve transport efficiency. Redesigned pallets and improved packaging systems, modeled after Returpack's batch return approach, could lower emissions and reduce logistical burdens (Löfgren et al., 2024). Additionally, optimizing crate designs to be lighter and more durable, as highlighted in interviews, could enhance sustainability throughout the supply chain.

Emerging tracking technologies, including QR codes, DataMatrix codes, and RFID as mentioned in interviews, offer solutions to improve bottle tracking, monitor lifecycles and monitor inventories as highlighted by to Rodrigues and Schilthuizen (Rodrigues, 2022; Schilthuizen, 1999). These tools enable detailed tracking of rotations, return rates, and bottle condition while providing transparency about sustainability initiatives to encourage consumer participation. However, scalability and cost-efficiency remain challenges. Collaborative efforts across the supply chain and modernization of Sweden's tracking infrastructure were noted as essential for successful implementation. Digitalization, such as automated order systems, could further optimize workflows and align operations with sustainability goals.

The PPWR explicitly encourages innovation, offering economic operators an additional five years to meet recyclability requirements for environmentally beneficial advancements (Council of the European Union, 2024).. This creates opportunities to adopt cutting-edge sorting technologies that enhance packaging system sustainability. Advanced recognition and material separation

capabilities, as emphasized in interviews, align with PPWR directives to improve efficiency and environmental performance.

7.1.7 Knowledge Sharing

The PPWR framework shares similarities with the Returpack and Returglas systems in its overarching goal of promoting sustainable packaging through efficient DRS management. However, the PPWR's EU-wide scope introduces opportunities for cross-border collaboration among Member States. This broader reach enables efficiencies, such as connecting DRS systems in neighboring regions to streamline transport and logistics. For example, border regions could share infrastructure, reducing redundancies and enhancing overall system efficiency.

The PPWR explicitly addresses these transboundary challenges by requiring Member States with significant cross-border business activity to ensure that DRS systems accommodate packaging from other countries (Council of the European Union, 2024). Additionally, it mandates mechanisms allowing consumers to reclaim deposits on packaging purchased in different Member States. These regulations underscore the importance of harmonization, ensuring sustainability efforts transcend national boundaries and contribute to a unified EU approach to achieving environmental goals.

Returpack's emphasis on knowledge sharing highlights the value of collaboration within DRS frameworks. Extending this approach to an EU context could improve transport efficiency and create synergies across systems. However, questions of ownership and economic responsibility arise. For instance, to equal the economic burden if a bottle purchased in Norway enters Sweden's system. A key consideration is whether achieving net-zero bottle exchange between countries justifies the administrative and logistical complexities involved. While economic factors will undoubtedly shape the decision-making process, the overarching objective of aligning with EU sustainability targets and tackling global environmental challenges could be taken in account. Balancing these economic considerations with sustainability objectives is essential for success. Stakeholders also pointed to the value of international collaboration where partnerships with which could facilitate knowledge sharing and operational improvements. Additionally, alignment with the EU's climate neutrality targets was identified as vital for ensuring long-term strategic relevance and fostering systemic progress.

The interviews also underscored the value of international collaboration. By learning from countries with established RGB systems, such as Germany, and sharing its own expertise, Sweden could enhance its recycling infrastructure. Stakeholders consistently noted that systemic changes, consumer engagement, and knowledge sharing are vital to overcoming challenges and achieving sustainable scaling of RGB systems. The comparison between CCEP SWE and CCEP DE underscores notable differences in beverage packaging sustainability. Germany's use of RGBs and rPET in a large scale highlights a well-established system with lower carbon intensity. As mentioned in the literature, 98% of packages are returned in Germany (Görgün, 2021). If Sweden would focus on expanding reusable packaging systems, streamlining logistics networks, and implementing comprehensive deposit-return schemes is essential. By analyzing best practices from Germany or other countries, and other countries Sweden could make significant strides in improving sustainability and reducing its carbon footprint.

7.1.8 Behavior and Communication

Findings from the interviews highlighted that the success of RGB systems relies on addressing consumer knowledge gaps, providing strong incentives, and ensuring convenience. These insights

align with a growing demand for products emphasizing social responsibility, safety, and health and for the company itself, it could help validate and improve circular products (Tabas *et al.*, 2024). The effectiveness of DRS in minimizing waste and encouraging reuse was strongly supported by interview findings. A dual strategy combining higher deposit values, convenient return systems, and targeted education is essential for overcoming barriers and validating the system's sustainability (Marampoutis et al., 2022).

Low return rates for consumers, identified as a recurring challenge in the interviews, highlight the need for targeted education and incentives. For instance, Systembolaget faces significant issues, with 86% of bottles being incorrectly discarded. On the other hand, the HoReCa sector is successful regarding the return rate of RGBs according to interviews, but having larger volumes of bottles could require efforts. Increasing deposit values, as suggested by interviewees, could positively influence consumer behavior, especially when combined with effective public awareness campaigns (Johansson & Henriksson, 2020). The interviews highlighted that many consumers remain unaware of RGB systems and their environmental benefits, underscoring the need for public awareness campaigns mandated by the PPWR. However, cultural barriers present additional challenges. Unlike Germany, where ingrained habits support sustainability, Sweden requires sustained efforts to embed these practices into daily life. This highly connected with the fact to overcome the obstacle of cultural factors including consumer interest and awareness which were highlighted to be a significant challenge in CE adoption and advancing CEE (Kirchherr et al., 2018; Tabas et al., 2024). The way of including the consumer in the circular business model is critical to not hinder the engagement. since consumers often lack both awareness and interest in circular offerings and their primary role could be to accept the new business model (Hobson & Lynch, 2016; Corvellec et al., 2022). It was further emphasized that consumers may be reluctant to return bottles it reused bottle appear less appealing, which further emphasize an awareness campaign on environmental benefits (Cinderby and McKendree, 2024).

The interviews revealed that effective consumer incentives are pivotal to promoting recycling behavior. Further highlighting the literature where a reward system incentivizes recycling behavior to encourage participation (Kim, 2021). Additionally, disparities in deposit values, such as 0.5 SEK for a 33 cl RGB versus 1 SEK for a small metal can or PET and 0.8 SEK per 50 cl RGB compared to 2 SEK per PET or metal can, were identified as influencing consumer choices, highlighting the need for better alignment between incentives and sustainability goals. This was further highlighted to combine the incentivization and education of consumer with for example engagement platforms with loyalty rewards, since educated consumer are more likely to participate in sustainable practices and therefore contributing to the system (Hobson and Lynch, 2016). Increasing deposits and could boost engagement for the targeted groups as discussed in the interviews and further influence purchasing behavior m (Johansson and Henriksson, 2020)

Accessibility also emerged as a critical factor. Interviewees emphasized that providing sufficient return locations, even in remote areas, significantly reduces barriers to recycling. Community influence was also noted as a driver, with evidence suggesting that individuals are more likely to recycle if their neighbors participate (Hage, 2009). These findings underscore the importance of combining improved accessibility with strategic incentives.

7.1.9 Market

According to the RISE LCA study, RGBs experienced a 3.8% decrease for 50 cl bottles and a 41.5% reduction for 33 cl bottles during this period (Löfgren et al., 2024). In contrast, NRGBs saw substantial growth, with 33 cl bottles increasing by 7.9% and 50 cl bottles by 3851%. This shift highlights a growing preference for NRGBs, contributing to a 9.8% overall increase in total glass

consumption Such trends raise concerns about Sweden's ability to meet long-term sustainability objectives. The growth of 50 cl NRGB bottles reflects preferences for convenience and variety but poses sustainability challenges due to higher carbon footprints. Stakeholder interviews revealed that while NRGBs are perceived as easy and circular. Glass's recyclability aligns with circular economy principles (Svensk Glasåtervinning, 2024).

The PPWR reuse targets provide a framework to reverse the decline in RGB usage by promoting a shift from NRGBs to RGBs (Council of the European Union, 2024). However, it remains crucial to understand whether consumer preferences for NRGBs are driven by their convenience or their non-refillable nature. Designing solutions that align with market demands while prioritizing reusable formats could help the industry leverage the growth in total glass consumption to meet sustainability objectives. This dual approach supports compliance with EU regulations and addresses increasing consumer demand for sustainable packaging.

Smaller bottle formats, such as 33 cl, require more bottles to deliver the same total volume as larger formats as seen in the analysis. RGBs, by reducing logistical demands like transport frequency and emissions, present a more sustainable alternative. Increasing RGB adoption in smaller formats could significantly reduce reliance on NRGBs, particularly in single-use-dominated categories. Additionally, focusing on the 50 cl format, where RGBs hold a larger share, offers further potential for optimizing logistics and reuse systems.

Despite these advantages, RGBs face challenges due to their heavier weight, a result of the sturdier materials needed for multiple reuse cycles. This weight disparity increases transport emissions and may deter consumers who prioritize lightweight and convenient packaging. Recent innovations, such as thinner and lighter bottle designs (Vinci, 2019; Löfgren et al., 2024), offer promising solutions to reduce the environmental footprint of RGBs while preserving their durability. Optimizing secondary packaging, such as modernizing crates or reducing their weight, could further enhance the sustainability and efficiency of RGB systems.

Barriers to RGB adoption include underdeveloped reverse logistics, insufficient economic incentives, and the convenience of single-use bottles. RGB systems in Sweden declined from a 61% market share in 1976 to under 2% in 2024, primarily due to logistical inefficiencies, outdated infrastructure, and shifts in consumer behavior (Johansson, 2022). Reintroducing RGBs requires addressing these historical challenges through collaboration, innovation, and strategic investment.

7.2 Conclusion

This research investigated the feasibility of implementing an RGB system in Sweden, focusing on two critical questions: *What strategies can be employed to maximize return rates of bottles in a RGB system and what are the strategic connections of implementing a RGB system at CCEP?* Given that CCEP Sweden does not currently operate an RGB system and is not directly affected by return rates, the study examined the changes required and its potential for alignment with CCEP's sustainability goals and compliance with the PPWR.

Coca-Cola's strong global brand identity is built on its iconic designs and extensive market presence. Transitioning to RGBs presents both significant challenges and transformative opportunities. This shift requires balancing innovation and adaptability to maintain brand consistency within a standardized framework. Simultaneously, it offers CCEP a unique opportunity to strengthen its leadership in environmental stewardship while leveraging sustainability as a competitive advantage. By embracing innovation and collaboration, the

company can drive a consumer-centered, environmentally sustainable business model that aligns with global trends.

The environmental benefits of transitioning to RGBs are substantial according to the interviews and analysis. Industry-wide, such a shift could reduce CO_2 emissions by 65%. with significant reductions in emissions for 33 cl and 50 cl bottles, at 69% and 54% respectively. Maintaining or extending the lifespan of RGBs, such as increasing the durability of 50 cl bottles beyond 8 cycles and 33 cl bottles beyond 40 cycles is essential for maximizing their sustainability potential. The transition to RGBs also aligns strategically with CCEP's net-zero GHG emission targets and supports The TCCC "World Without Waste" initiative, which aims to achieve 25% of global beverage volume in refillable formats. This alignment reinforces the importance of sustainability as both a corporate priority and a competitive differentiator.

The analysis of the interviews shows that transitioning operationally to RGBs requires investments in supply chain, integrating forward supply chain with reverse supply chain in a closed loop supply chain, with infrastructure and storage, reverse logistics systems, and consumer-centric processes. Learning from best practices from CCEP Germany provides insights into scalable infrastructure, consumer-friendly practices and successful implementation in Sweden at a larger scale. Sustainable transport solutions, such as the use of HVO100 fuel or electric vehicles, and efforts to reduce bottle weight, could further minimize the system's environmental impact. Establishing and monitoring key performance indicators (KPIs) is essential for tracking progress, identifying challenges, and ensuring continuous improvement.

Maximizing return rates necessitates strategies such as optimizing the current DRS, implementing targeted educational campaigns, and offering tailored incentive programs to encourage participation, especially for consumers. Accessibility and convenience, such as strategically placing RVMs and ensuring adequate coverage in suburban and rural areas, are vital for fostering consistent consumer behavior. Engaging platforms with loyalty rewards and incentives could further encourage recycling participation, while public awareness campaigns could bridge knowledge gaps regarding the environmental benefits of RGBs. Addressing cultural barriers and embedding sustainable practices into daily life will require sustained efforts, including consumer engagement initiatives and collaboration. The research highlights the critical role of multi-stakeholder collaboration, including producers, municipalities, retailers, and governing bodies. Effective models like Returpack demonstrate the potential of shared accountability and regulatory alignment in achieving high return rates. Additionally, cross-border collaboration could harmonize deposit-return systems across EU Member States, enabling shared infrastructure, reducing redundancies, and fostering knowledge exchange.

This research contributes academically by integrating environmental sustainability, consumer behavior, and operational logistics to create a comprehensive framework for corporate sustainability strategies in packaging transitions. It demonstrates that RGB systems are not only critical for advancing circular economy principles and waste management but also for aligning corporate goals with global sustainability trends. Moreover, the research underscores how regulation-driven innovation, such as compliance with the PPWR, could drive the circular economy, offering both economic and reputational advantages to early adopters. CCEP Sweden's transition to RGBs could serve as a benchmark for the beverage industry, promoting best practices and driving industry-wide adoption of circular economy principles.

From another academic perspective, this research contributes to the growing body of knowledge on circular economy practices and sustainable packaging. It bridges the gap between theory and practice by offering actionable insights into implementing RGB systems in a national context, with potential applications in similar markets. This thesis addresses a critical gap in understanding the

feasibility and strategic integration of RGB systems in Sweden. By combining lifecycle assessments, regulatory insights, and industry data, it provides a comprehensive evaluation of how refillable systems could drive sustainable innovation and operational efficiency. The study examines strategic alignment with the EU Packaging and Packaging Waste Directive (PPWR), operational feasibility, and the broader implications for the beverage sector. Furthermore, it advances the understanding of lifecycle impacts and operational strategies for transitioning to refillable systems, highlighting their role in achieving both national and global sustainability goals.

In conclusion, transitioning to RGBs represents a pivotal step toward sustainable innovation for CCEP. While the initial investments, including infrastructure development and reverse logistics, may seem significant, the long-term economic benefits, such as reduced material costs, enhanced brand loyalty, and compliance with sustainability regulations, provide compelling justification for this initiative. Success will depend on industry-wide collaboration, supportive regulatory policies, and proactive consumer engagement. By embracing this transformation, CCEP could lead the beverage industry in setting a global standard for sustainability. This initiative not only reinforces CCEP's commitment to environmental stewardship but also demonstrates that innovation can drive a more sustainable future. As the industry navigates evolving consumer expectations and regulatory landscapes, CCEP's leadership in transitioning to RGB systems could pave the way for a more sustainable and circular economy.

8 RECOMMENDATIONS AND FUTURE WORK

This section provides recommendations for CCEP, emphasizing strategies to maximize return rates, ensure regulatory compliance, and achieve sustainability goals, with the broader industry's role identified as critical to success. These recommendations are grounded in insights from literature, quantitative and secondary data, and comprehensive analysis, ensuring they are practical and well-informed. Further investigation could focus on consumer behavior, advanced technologies, economic feasibility, and scalable logistics to address key gaps and support the long-term sustainability and effectiveness of RGB systems.

8.1 For CCEP

Based on the insights from the research, for CCEP to fully leverage the potential of the RGB system, internal collaboration across departments and operational alignment will be important. Implementing an efficient and scalable RGB system requires seamless coordination between production, logistics, sustainability, marketing, and regulatory teams.

8.1.1 Modify or Establish a New RGB Production Line

A RGB production line at CCEP was removed and replaced by the NRGB production line as highlighted in the interviews. Based on insights from CCEP SWE and CCEP DE logistics experts the following steps were concretized to modify its existing production line or establish a new one to accommodate essential steps for processing RGBs. The following key steps were recommended:

- **Crate Inspection:** Establish a process to inspect the structural integrity of crates used for transporting and storing returned bottles, ensuring they meet operational requirements.
- **Case Unpacker:** Automate the unpacking process to efficiently separate bottles from crates, reducing manual handling and increasing throughput.
- **Inspection Station:** Add an inspection stage equipped with detection systems to identify and remove damaged, contaminated, or unsuitable bottles.
- **Bottle Washer:** Introduce an industrial washing system capable of thoroughly cleaning returned bottles to meet hygiene and quality standards.
- **Sorting Station:** Implement automated sorting mechanisms to categorize bottles by type, size, or color, ensuring compatibility with subsequent refilling processes.

As mentioned in the interviews the old RGB system was exchanged due to inefficiencies and therefore a key aspect would be to optimize the processes. Incorporating advanced cleaning and inspection technologies could enhance operational efficiency, maintain quality, and reduce bottlenecks in the system, while saving water and energy. The space required for the RGB production line within the facility needs to be evaluated as highlighted in the interviews. Adequate spatial planning ensures that the new or modified production line integrates seamlessly with existing operations while accommodating future scalability and efficiency improvements.

Additionally, interviews highlighted the potential to integrate digital markers. Scannable identifiers could be directly integrated into the bottle design or attached via durable labels and should be easy to remove during the washing process or incorporate a design that does not require removal. This approach ensures seamless tracking and operational alignment without compromising bottle aesthetics or sustainability goals. Such innovations could reduce washing time, improve traceability, and simplify the overall process.

RGBs were evidently lower in the carbon footprint comparing to NRGBs from the carbon footprint analysis, where the aspect of maximizing bottle lifecycle was important to minimize carbon footprint. The design of the bottles themselves should focus on sustainability and durability while preserving aesthetic appeal. Bottles should be lightweight and space-efficient to optimize transport and storage, utilize recycled or sustainable materials to reduce environmental impact, and be durable enough to withstand repeated washing and refilling cycles. Furthermore, validation of bottle lifespans and extending it would be beneficial. This approach ensures the bottles are both practical and environmentally responsible.

CCEP could also benefit from benchmarking against best practices for RGB systems in regions such as Germany, to compare to larger scale operating RGB systems. Learning from the successes of other markets could help CCEP identify and adopt sustainable practices, improve operational efficiency, and enhance technological integration.

8.1.2 Establishing an Efficient Closed-Loop Supply Chain

The success of an RGB system depends on establishing an efficient closed-loop supply chain that ensures seamless collection, storage, and redistribution of bottles for reuse as noted in interviews. Collaboration with key stakeholders, including wholesalers, retailers, breweries, Returglas, customers, and consumers, are essential for smooth operations and maximizing return rates. Stakeholders should share responsibilities and benefits, while wholesalers and retailers could serve as central collection points to streamline the system.

While interviews indicate that the RGB system is already well-functioning within the HoReCa sector regarding bottle return rates, a significant gap remains in the end-user and consumer segment regarding participation and consumer knowledge. Furthermore, one key limitation identified was that RVMs, which are part of the existing DRS, could be insufficient in handling increasing volumes of RGBs, meaning that an expansion of infrastructure could be necessary to ensure efficient handling of RGBs. To address these insights, CCEP could explore the establishment of decentralized refill hubs that allow consumers and customers to return and refill bottles more conveniently at a smaller scale. For the HoReCa sector, a refill hub on-site, could reduce the need for extensive return logistics while maintaining brand consistency and quality control. Instead of relying solely on a centralized facility in Jordbro, a network of smaller refill hubs distributed across strategic locations could improve accessibility, reduce transportation distances, lower logistics costs, and decrease carbon emissions. By processing returned bottles closer to their point of use, these hubs would not only improve bottle availability but also create a more flexible and responsive supply chain. To further engage consumers, smart return stations could be equipped with tracking technologies that could incentivize the consumers.

In an RGB system, prioritizing accessible and consumer-friendly collection networks is essential to encourage active participation, as identified in the interviews. Strategically located collection points, designed to reach consumers effectively, could enhance convenience and provide incentives to improve return rates. This could not only strengthen consumer trust but also integrate sustainability into daily routines. To address geographical differences, regional tailoring of logistics strategies is crucial. The current RVM network for RGBs may be insufficient if RGB volumes increase, requiring an optimized distribution of collection points. Urban areas, such as Stockholm could face challenges due to high demand, while suburban and rural regions need innovative solutions to ensure accessibility and efficiency. Additionally, seasonal variations, highlighted in the interviews, should be accounted for to maintain consistent system performance throughout the year. A well-balanced RVM network is vital to supporting Sweden’s long-term sustainability goals.

As further mentioned in the interviews, the storage of an RGB system, where empty bottles and heavier weight takes up more space than NRGBs. Efficient storage management is necessary to maintain organization and readiness for reintegration into the production cycle. Bottles could be categorized by type, size, and condition to streamline subsequent processes, with real-time inventory systems in place to prevent delays and ensure proper handling. These practices minimize damage and optimize bottle reuse.

As seen in the RISE LCA study analysis and from the interviews, sustainable fuel such as HVO and electric vehicles could to further lower the carbon footprint. The use of renewable energy and could furthermore support long-term sustainability goals. The load capacity and challenges with transporting RGBs were noticed in the interviews. Optimizing logistics could be critical to optimize transportation routes, load consolidation, reducing costs, and enhancing overall system sustainability. Predictive analytics and real-time data integration could improve supply chain coordination and vehicle loading efficiency while reducing emissions through minimized empty trips. Mentioned in the interview of a closed-loop system, combining forward and reverse logistics into a unified system could maximize operational efficiency and reduce environmental impact. Continuous improvement could be supported by key performance indicators (KPIs) to monitor system effectiveness. Metrics such as return rates, transportation emissions, and collection efficiency will enable data-driven decisions and strategic adjustments, ensuring the system adapts to emerging challenges and opportunities.

8.1.3 Consumer Engagement and Incentives

Based on the insights from the interviews, engaging consumers is a pivotal component in optimizing the RGB system, requiring a comprehensive strategy that fosters active participation, long-term commitment, and trust. CCEP should prioritize initiatives that motivate consumers while building meaningful connections through incentives, digital platforms, gamification, and sustainability messaging. Introducing loyalty programs could reward consistent bottle returns with tangible benefits such as points, discounts, or cashback could effectively encourage participation while simultaneously cultivating brand loyalty. For socially conscious individuals, offering the option to donate deposit refunds to charities or sustainability initiatives could amplify engagement, allowing consumers to connect their actions with broader societal and environmental goals. This dual focus on personal rewards and social impact strengthens trust and ensures long-term commitment to the RGB system.

Convenience plays a role in driving return rates. The RVMs should ensure that collection points feature user-friendly interfaces and clear instructions, removing barriers to participation and enhancing consumer satisfaction. Furthermore, a digital platform, such as a mobile application, could enhance user experience and accessibility. The platform could guide consumers to nearby RVMs or collection points, track their contributions, and visually display their cumulative environmental impact, such as reductions in waste and carbon emissions. Providing real-time feedback on individual efforts reinforces the connection between personal actions and sustainability outcomes, simplifying interactions and encouraging greater participation. Gamification strategies could further transform the return process into an engaging and enjoyable activity. Digital challenges, leaderboards, badges, and milestone rewards could introduce elements of fun and competition to routine tasks. For instance, challenges might incentivize users to meet return targets within a set timeframe, while leaderboards could foster a sense of community. Achievement systems could reward consumers for consistent participation or reaching specific milestones, motivating sustained engagement.

Interviews indicated that customers lacked the knowledge of the benefits of a RGB system. Educating consumers about the environmental benefits of returning bottles could therefore be

critical for fostering a sense of purpose and responsibility, changing a behavior. Clear and engaging messaging could highlight key impacts, such as reduced carbon emissions, minimized material waste, and contributions to circular economy principles. Leveraging social media, in-store promotions, and targeted campaigns to communicate these benefits effectively could maximize outreach and create a stronger connection between consumers and the RGB system. Linking bottle returns to meaningful causes, such as supporting community projects or renewable energy initiatives, could inspire greater consumer engagement. By aligning personal actions with broader social and environmental goals, CCEP could foster an emotional connection between the RGB system and its users. Providing relatable and tangible information about how individual returns contribute to broader sustainability goals reinforces consumer motivation and builds trust in the system.

8.1.4 Leveraging Innovative Technologies

As mentioned in interviews, technologies to improve a RGB system exists and are under investigation. Implementing technological advancements could be an opportunity for enhancing the efficiency, sustainability, and effectiveness of the RGB system. By integrating advanced tools across the production line and supply chain, CCEP could streamline operations, improve transparency, and optimize consumer engagement while minimizing environmental impact.

The implementation of a RGB production line involves critical activities such as inspection, unpacking, washing, and sorting of returned bottles where advanced technologies could be leveraged to optimize these processes. AI-powered inspection systems could accurately and rapidly identify and remove damaged, contaminated, or unsuitable bottles, minimizing errors and ensuring high-quality standards. Similarly, advanced sorting mechanisms could streamline bottle categorization by type, size, or color, reducing bottlenecks and enhancing operational efficiency. Innovative cleaning technologies could minimize water and energy consumption during the washing process. These innovations align with sustainability objectives by reducing the production line's environmental footprint while maintaining performance and hygiene standards.

Digital markers could enable precise traceability of bottles and ensure transparency across all stages of the supply chain. These digital markers could improve traceability, facilitate inventory management, and support data-driven decision-making across the supply chain. Additionally, such systems could identify logistical bottlenecks and optimize process efficiency, ensuring smooth operations within the closed-loop system. Predictive analytics powered by AI could forecast demand, optimize transportation routes, and reduce inefficiencies, ensuring a seamless flow of bottles within the closed-loop system. Automated logistics platforms further enhance coordination by optimizing vehicle loads, reducing empty trips, and minimizing emissions, thus improving the environmental performance of the entire system.

RVMs play a pivotal role in improving return rates by providing consumers with a seamless and accessible recycling solution. Expanding and managing RVM to be deployed at strategic locations such as supermarkets and transit hubs could increase accessibility and participation. Innovations in RVM technology, including faster processing speeds, multi-format acceptance, and user-friendly interfaces, enhance the consumer experience. Smart RVMs equipped with AI capabilities and real-time data integration could optimize collection schedules, monitor return volumes, and streamline logistics.

8.1.5 Positioning CCEP as a Sustainability Leader

From the analysis of the PPWR and alignment with the business perspective, the RGB system presents a significant opportunity. CCEP could establish itself as a leader in sustainable packaging innovation and enhance brand reputation and strengthen its competitive advantage within the beverage industry. This approach aligns with the company's broader business objectives, which emphasize market leadership, operational innovation, and consumer engagement. From a strategic perspective, leveraging the RGB system as a platform for innovation enables CCEP to pioneer sustainable packaging solutions and redefine industry standards. This initiative also provides an avenue to strengthen consumer trust and loyalty by demonstrating a strong commitment to environmental stewardship while delivering a seamless and user-friendly return process.

The alignment of the RGB system with sustainability goals efforts further underscores its strategic importance. By integrating refillable bottles into its operations, CCEP can drive innovation in sustainable packaging and visibly contribute to global environmental objectives. This initiative also aligns with circular economy principles by promoting material reuse, minimizing waste, and reducing reliance on virgin resources. Furthermore, the RGB system should align with the PPWR, addressing current compliance requirements while anticipating future updates. Collaborating with policymakers and industry peers will enable CCEP to proactively shape and adapt to regulatory changes, reinforcing its position as a leader in sustainable packaging.

To measure and optimize the system's environmental impact, CCEP should adopt key sustainability metrics. LCA could provide further critical insights into the benefits of RGBs, including reductions in carbon emissions, energy consumption, and waste generation. Additionally, reaching data-driven targets, such as increasing return rates and lowering the carbon footprint, will allow CCEP to track progress and refine strategies over time. By embedding the RGB system within a circular economy framework, CCEP can maximize resource efficiency and ensure long-term sustainability.

8.2 For the Industry

Based on the insights from the results, a successful implementation of RGB systems depends on collaboration across the beverage industry to establish standardized practices, shared infrastructure, and streamlined operations. Industry-wide efforts should focus on clear roles, robust incentives, and consumer engagement to drive participation and create scalable solutions.

8.2.1 Collaboration in the Industry: A Foundation for Success

Collaboration among key stakeholders is essential for the successful implementation of an RGB system as highlighted from the interviews insights of internal and external stakeholders. Beverage producers are central to this process, responsible for designing and developing RGB systems, ensuring regulatory compliance, and making operational adjustments. Retailers and wholesalers facilitate the system by providing accessible collection points and participating in initiatives that encourage consumer engagement.

Policymakers and regulators play a critical role in establishing regulatory frameworks and offering targeted incentives to accelerate the adoption of RGB systems as seen in the PPWR. Collaboration between policymakers, producers, retailers, and wholesalers ensures the creation of flexible yet effective frameworks that meet regulatory goals while fostering innovation and adaptability to changing market demands. This coordinated approach lays a strong foundation for effective and widespread implementation.

Efficient collection networks are a crucial element of collaboration. These networks should prioritize consumer convenience, ensuring bottle returns are accessible in both urban and rural areas. Retailers and wholesalers should be incentivized to establish collection points such as RVMs while advanced logistics tools optimize routes, reduce costs, and minimize emissions. A well-structured and collaborative network enhances system efficiency, builds trust, and drives participation across the value chain. By aligning efforts, stakeholders can create a robust and sustainable system that achieves environmental and operational objectives.

8.2.2 PPWR as a Catalyst for Sustainable Innovation

The PPWR should be embraced as a driving force for sustainable innovation. This regulatory framework goes beyond compliance requirements by offering valuable guidance for overcoming challenges in reusable packaging systems. By aligning industry practices with PPWR directives, companies can standardize processes, foster innovation, and accelerate progress toward sustainability goals. Furthermore, the PPWR could serve as a strategic roadmap for addressing critical issues such as waste reduction, resource efficiency, and the integration of circular economy principles. It empowers stakeholders to develop scalable solutions that address both operational and environmental challenges. By leveraging the regulation's objectives, the industry can design systems that are compliant, forward-thinking, and adaptable to evolving market demands.

Adopting the PPWR positions the industry as a leader in sustainability and circular economy initiatives. Beyond meeting regulatory obligations, businesses can use the framework to proactively address environmental challenges, improve operational efficiency, and strengthen consumer trust. By viewing PPWR compliance as an opportunity, companies can transform requirements into a platform for innovation and competitive growth.

To fully capitalize on the PPWR the reuse systems should meet current mandates and anticipate future updates. Collaboration with policymakers is crucial for aligning regulatory goals, gaining insights into upcoming changes, and contributing to the creation of supportive and flexible frameworks. By integrating these strategies, the industry can turn the PPWR into a cornerstone for sustainable development and long-term success.

8.2.3 Comprehensive Incentive Strategies Across the Value Chain

Incentives are pivotal in driving engagement and participation across the value chain, addressing financial, behavioral, social, technological, and environmental dimensions. A well-structured incentive model ensures active stakeholder contribution and supports the success of the RGB while driving participation, foster innovation, and achieve long-term success. This strategy ensures alignment between stakeholder engagement and sustainability goals, creating a robust and impactful system.

Financial incentives are foundational for encouraging returns as mentioned in the interviews and from the theoretical background. The return symbol placed on the bottles could be more effective in the communication about the incentives to participate in the RGB system. Furthermore, increasing deposit values could motivate consistent bottle returns, while bulk bonuses for large quantities incentivize higher participation rates. Partnering with retailers to offer loyalty rewards, such as converting refunds into store credits or discounts, provides additional motivation and fosters long-term consumer loyalty.

Behavioral incentives make the return process engaging and enjoyable where loyalty rewards, such

as discounts or redeemable points at retailers, encourage consistent participation. Gamification strategies, including mobile apps and interactive RVMs with features like challenges, leaderboards, and achievements, make the return process enjoyable and rewarding. Offering consumers the option to donate refunds to charities or environmental initiatives introduces a meaningful social impact element, appealing to socially conscious individuals and enhancing their commitment to the system. Social incentives connect consumer actions with meaningful causes. Offering the option to donate refunds to charities or environmental initiatives appeals to socially conscious individuals, creating a dual impact by fostering emotional connections with the RGB system and supporting broader social goals.

Technological advancements streamline the return process and enhance accessibility. Expanding the availability of RVMs at strategic locations increases convenience, while mobile applications guide users through the return process, track their contributions, and display real-time environmental impact. These tools simplify logistics and enhance the overall user experience.

Environmental incentives connect consumer actions to measurable sustainability outcomes. Highlighting the ecological benefits of RGB systems, such as reduced carbon emissions, minimized waste, and improved resource efficiency, through targeted campaigns significantly enhances trust and commitment. Sustainability branding could further amplify consumer engagement by linking individual efforts to tangible environmental benefits. By demonstrating how consumer actions contribute to broader sustainability goals, branding fosters trust, loyalty, and long-term participation. This approach reinforces the industry's commitment to the circular economy and environmental stewardship.

8.2.4 Education and Awareness for Driving Engagement

Overcoming psychological and habitual barriers to consumer participation in RGB systems in Sweden was highlighted as a challenge from the interviews. To address these challenges, sustained efforts are necessary to integrate reuse systems into daily routines and promote a cultural and behavioral shift toward sustainability by educating consumers on the RGB system and its incentives to have a widespread engagement. Education and awareness campaigns could simplify the return process by providing clear and actionable instructions where RGB system operates and how to participate effectively, furthermore improving system efficiency and maintain the quality of returned bottles. As highlighted in the interviews, specific gaps exist in consumer knowledge across different demographic groups. To maximize reach and impact, communication strategies should tailor messaging to targeted customer segments, particularly those with lower return rates. A combination of digital outreach, in-store promotions, and community-driven initiatives could ensure that education efforts are inclusive, engaging, and effective. Connecting individual actions to broader goals creates a tangible link between consumer behavior and meaningful sustainability outcomes.

8.2.5 Limitations and Future Work

This study provides a robust foundation for understanding the implementation of RGB systems. However, several limitations must be acknowledged, and they highlight important avenues for future research to enhance the scalability, effectiveness, and long-term success of RGB systems in diverse markets and regulatory environments.

One key limitation of the study lies in its reliance on secondary data and stakeholder interviews, which restricted direct insights into consumer behavior and localized challenges. Without

consumer surveys or region-specific pilot studies, the study could not fully capture the behavioral drivers or operational hurdles unique to different demographics and regions. The real-world testing leaves room for further work and validation.

Certain assumptions in the RISE LCA study also introduce uncertainties. For example, the estimated lifespans of 33 cl and 50 cl bottles at 40 and 8 cycles, respectively, may overstate the environmental benefits if actual lifespans are shorter. Similarly, transport distances were uniformly assumed to be 250 km one-way, neglecting variations in regional logistics and potential backhaul inefficiencies, which could distort emissions estimates. The presumption of a 100% collection rate for RGBs represents an ideal scenario. Furthermore, the allocation of 10% of brewery energy to bottle cleaning simplifies operational requirements where emissions would vary from these processes.

Although this study explored technological solutions such as AI, RFID, and advanced tracking systems, these were discussed conceptually, without empirical validation of their scalability or practical implementation. Stability in regulatory frameworks, such as the PPWR was assumed, but future changes could significantly influence the feasibility and design of RGB systems.

To address these limitations, future research could focus on several key areas. Direct consumer surveys and behavioral studies could be conducted to understanding the drivers, barriers, and trends influencing participation in RGB systems. Expanding these studies across diverse demographics, cultural contexts, and regional differences could provide actionable insights for designing effective incentives, educational campaigns, and engagement strategies. Region-specific pilot projects could validate these approaches, providing critical data on return rates and consumer acceptance. Strategic implementation of RGB systems could provide valuable data for refining strategies, addressing operational challenges, and evaluating consumer behavior. Insights gained from these pilots could inform the development of scalable solutions that balance sustainability and operational efficiency, supporting broader adoption.

Operational and logistical challenges could need further exploration. Region-specific analyses of logistics and collection systems could inform tailored implementation strategies. Advanced methodologies, such as route optimization, real-time data integration, and predictive analytics, could enhance reverse logistics, reduce costs, and minimize emissions. Developing integrated logistics platforms to optimize vehicle loads and reduce empty trips could further contribute to creating seamless, closed-loop systems. Future studies might also investigate the most suitable DRS model for Sweden to ensure system efficiency and scalability.

The scalability and cost-effectiveness of advanced technological solutions, such as AI-driven logistics and RFID tracking, should also be evaluated. Research and pilot projects could assess the practical applicability of these innovations under real-world conditions, focusing on their ability to improve transparency, lifecycle monitoring, and operational efficiency. Additionally, the development of centralized data management systems capable of handling large volumes of information would be crucial to the success of RGB systems.

Advanced LCAs could expand beyond carbon emissions to address other environmental factors, including water usage. Comparative studies of RGBs and alternative packaging options, such as refillable PET, aluminum cans, and other solutions and materials, could guide the development of sustainable packaging strategies tailored to specific market needs and goals. Detailed financial assessments, sensitive to market fluctuations and material costs, could complement these efforts by evaluating the economic feasibility of transitioning to RGB systems, factoring in logistics, infrastructure investments, and long-term viability.

Finally, global comparisons and best practices could provide valuable insights for region-specific implementation strategies. Studying RGB adoption in markets such as Germany could highlight effective approaches, while analyzing the challenges faced in less-developed systems could guide solutions for Sweden and beyond. By learning from these experiences, Sweden can adapt successful strategies to address its unique challenges. Cross-sector collaborations offer another promising investigation for advancing RGB systems where partnerships within the industry, governments, and logistics providers could leverage shared infrastructure and expertise to uncover cost-effective solutions. Such collaborations may accelerate the development of efficient reverse logistics systems and foster innovation.

By integrating these research directions, the beverage industry could develop a RGB model that advance the transition to a circular economy. Combining innovative packaging solutions, advanced logistics systems, and comprehensive consumer insights could allow organizations to address environmental challenges more effectively. Tailored strategies that technologies and foster collaborative frameworks could drive significant reductions in waste, improve resource efficiency, and minimize emissions. These advancements could reinforce the industry's commitment to sustainability while positioning companies like CCEP as global leaders in environmental stewardship to meet the demands of a changing world.

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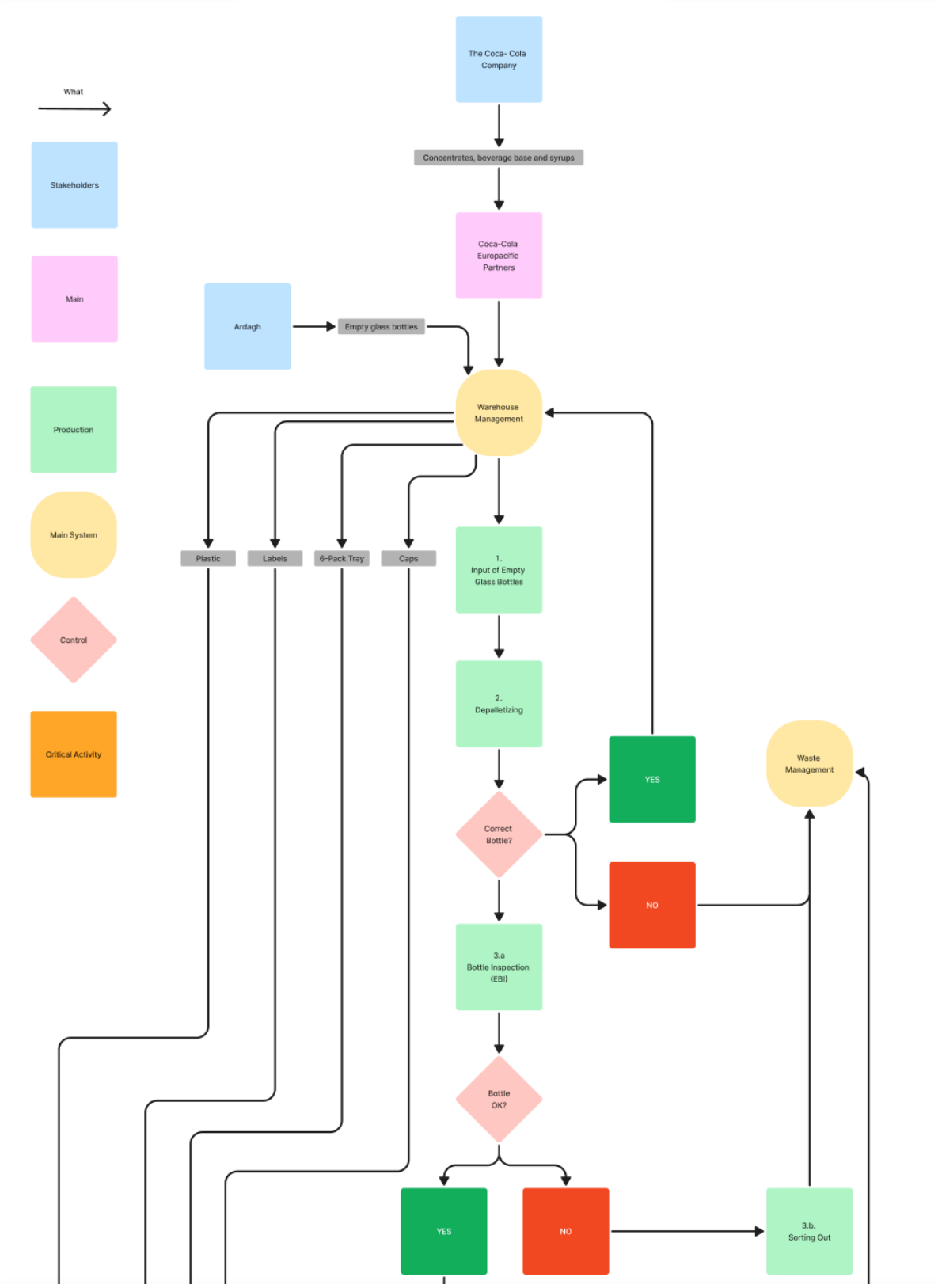
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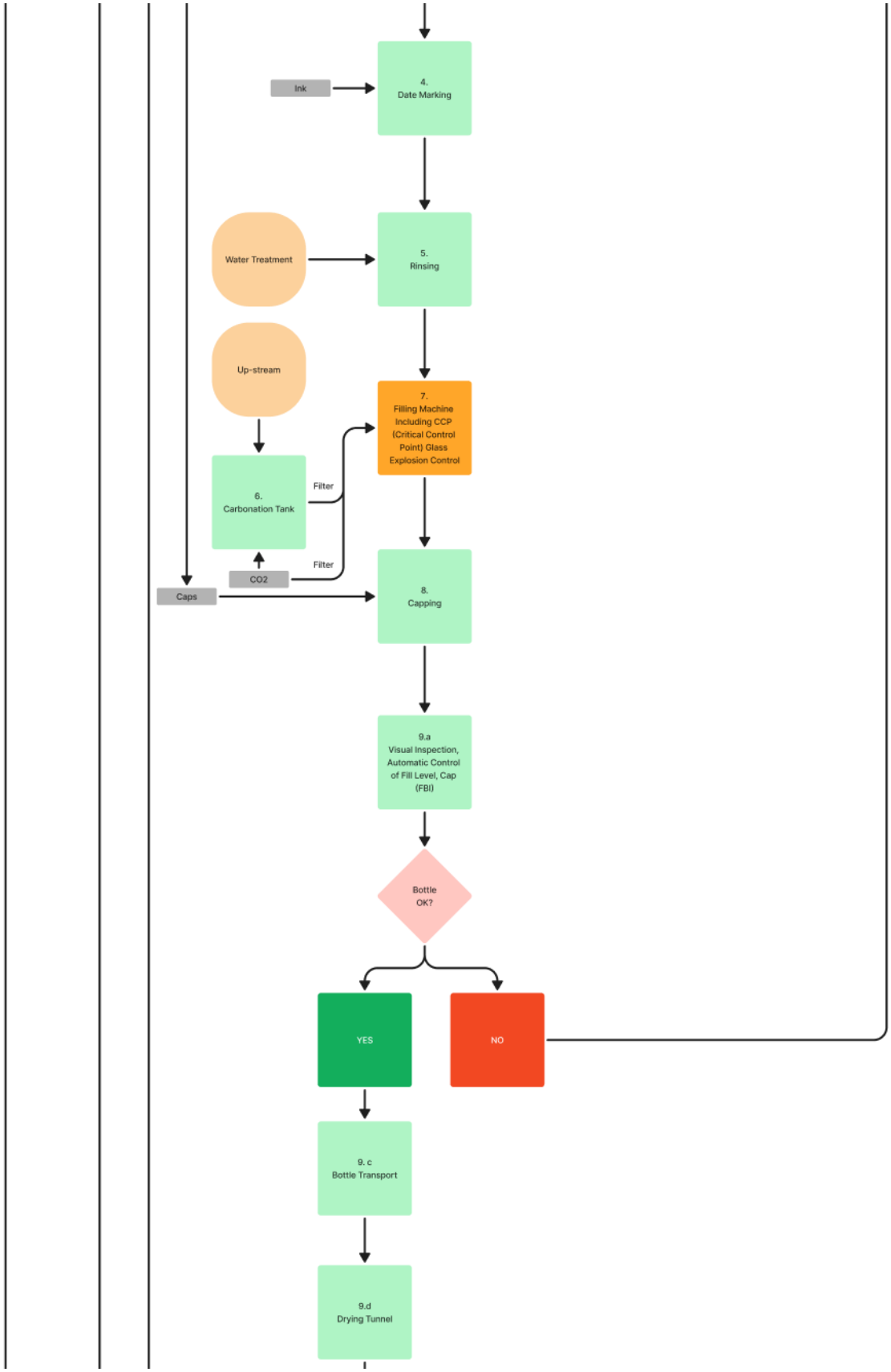
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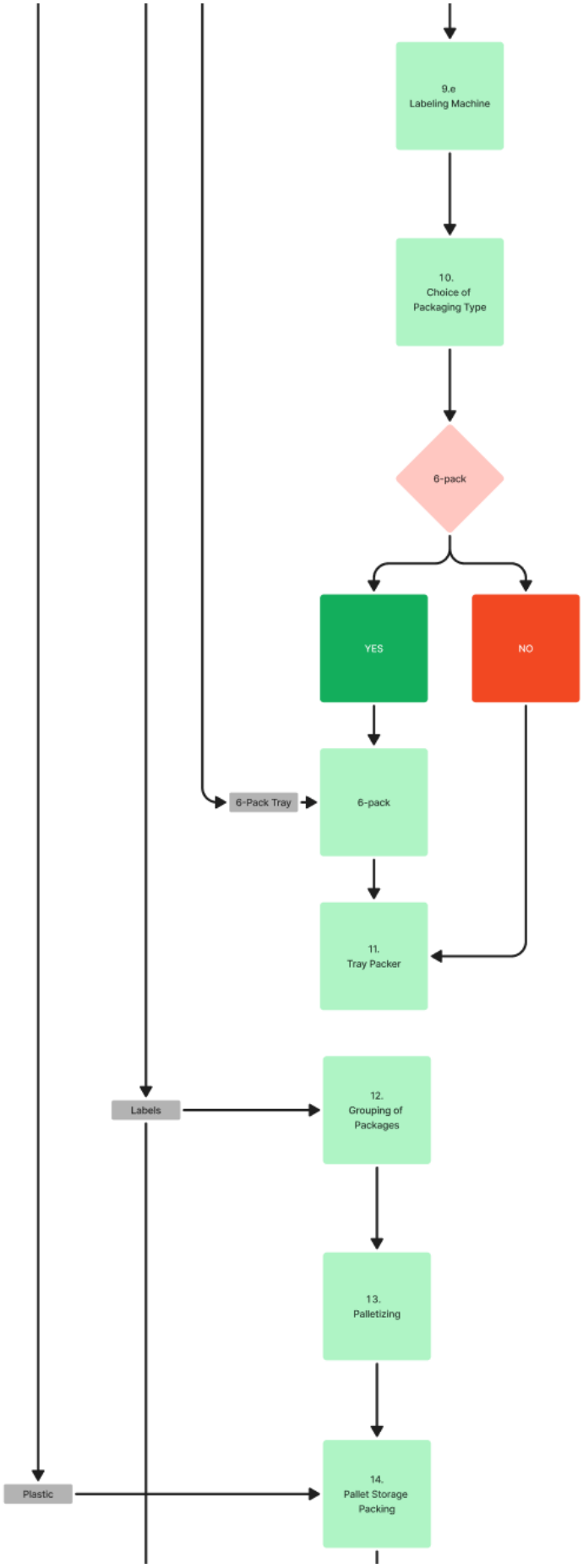
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APPENDIX A: CURRENT GLASS PRODUCTION LINE

To investigate which activities needed to be analyzed and modified to have a RGB system, a modelling of the current glass production line was made, as seen in Figure A1.




Ink
4.
Date Marking
Water Treatment
5.
Rinsing
Up-stream
7.
Filling Machine
Including CCP
(Critical Control
Point) Glass
Explosion Control
6.
Carbonation Tank
Filter
Filter
CO2
Caps
8.
Capping
9.a
Visual Inspection,
Automatic Control
of Fill Level, Cap
(FBI)
Bottle
OK?
YES
NO
9. c
Bottle Transport
9.d
Drying Tunnel


9.e
Labeling Machine
10.
Choice of
Packaging Type
6-pack
YES
NO
6-Pack Tray
6-pack
11.
Tray Packer
Labels
12.
Grouping of
Packages
13.
Palletizing
Plastic
14.
Pallet Storage
Packing

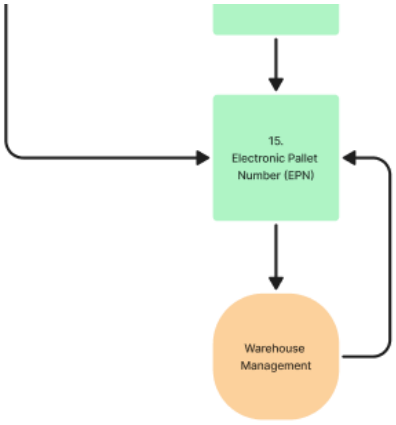


Figure A1. NRGB Production Line

APPENDIX B: REFILLABLE GLASS BOTTLE PRODUCTION LINE

The implementation of a RGB production line introduced several new activities, including the use of machines for inspection, unpacking, washing, and sorting bottles. These activities, as illustrated in Figure B1, form the backbone of the added production process for a RGB production line, ensuring that returned bottles are efficiently prepared for reuse.

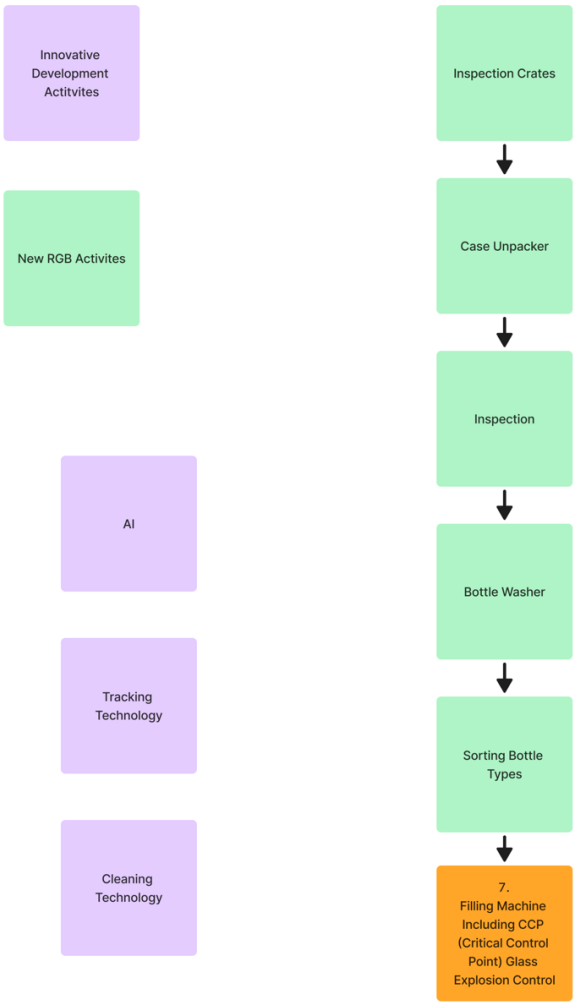


Figure B1. RGB Production Line New Activities

To enhance these processes, advanced technologies can be integrated. For example, artificial intelligence (AI) and advanced sorting mechanisms could be used to improve the precision and speed of bottle inspections, ensuring that damaged or unsuitable bottles are identified with minimal error. Tracking technologies could streamline logistical operations, enabling more effective monitoring of bottle movement and inventory management throughout the production line.

Innovations in cleaning technology also offer opportunities for greater sustainability. By adopting techniques that minimize water and energy usage, such as ultrasonic cleaning or advanced filtration systems, the environmental impact of the washing process can be significantly reduced. These innovations not only improve the efficiency of the production line but also align with broader sustainability goals.

APPENDIX C: REFILLABLE GLASS BOTTLE DISTRIBUTION SYSTEM

The RGB distribution system is shown in Figure C1 where necessary activities added for a RGB system compared to NRGBs are highlighted in red.

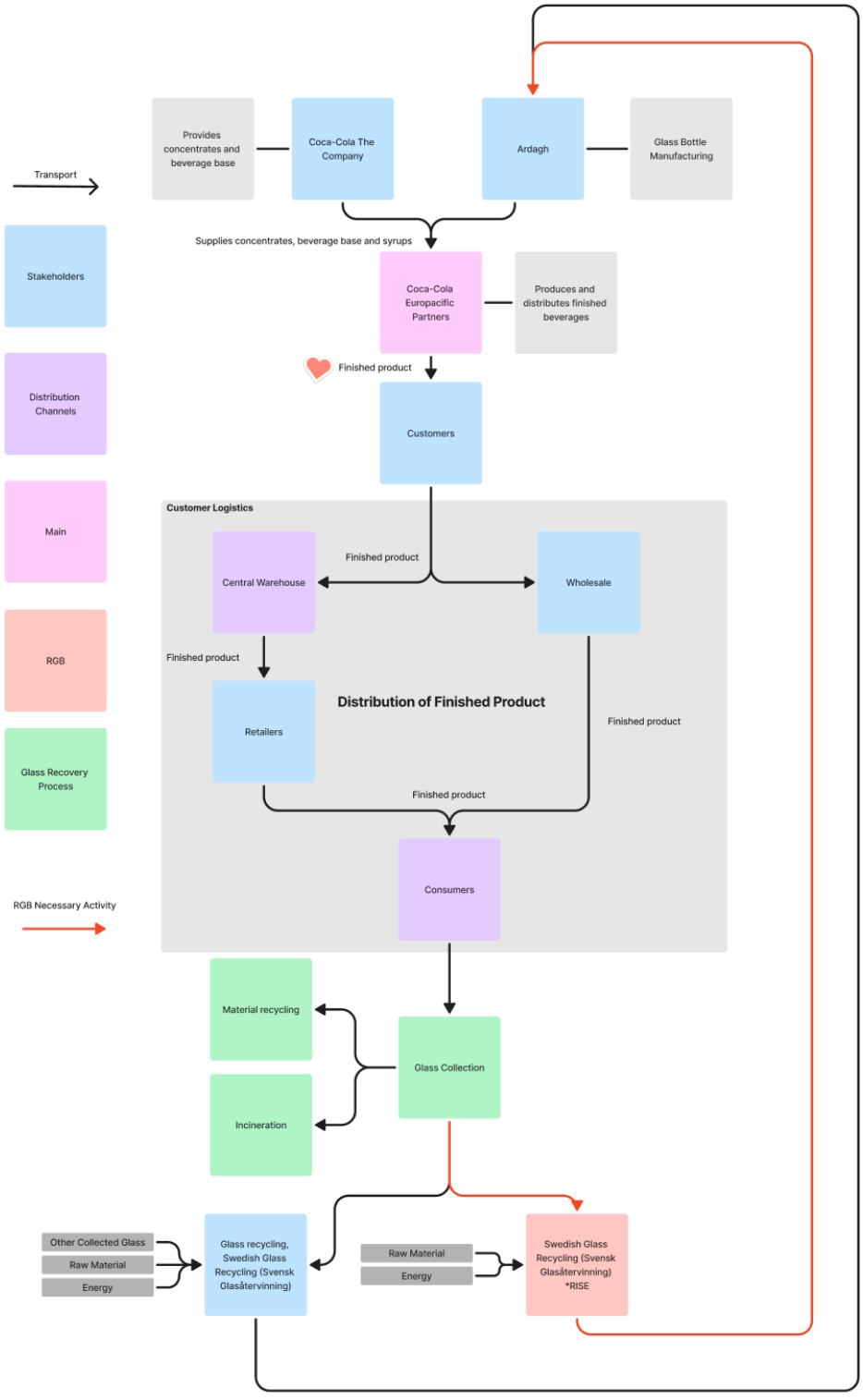


Figure C1. Distribution System

APPENDIX D: REFILLABLE GLASS BOTTLE CLOSED-LOOP SYSTEM

The extended customer logistics for RGB should also include reverse logistics activities to have a closed-loop system, as illustrated in Figure D1. These activities could include management of storing and route optimization. The breweries, consumers, Returglas and costumers are the key stakeholders..

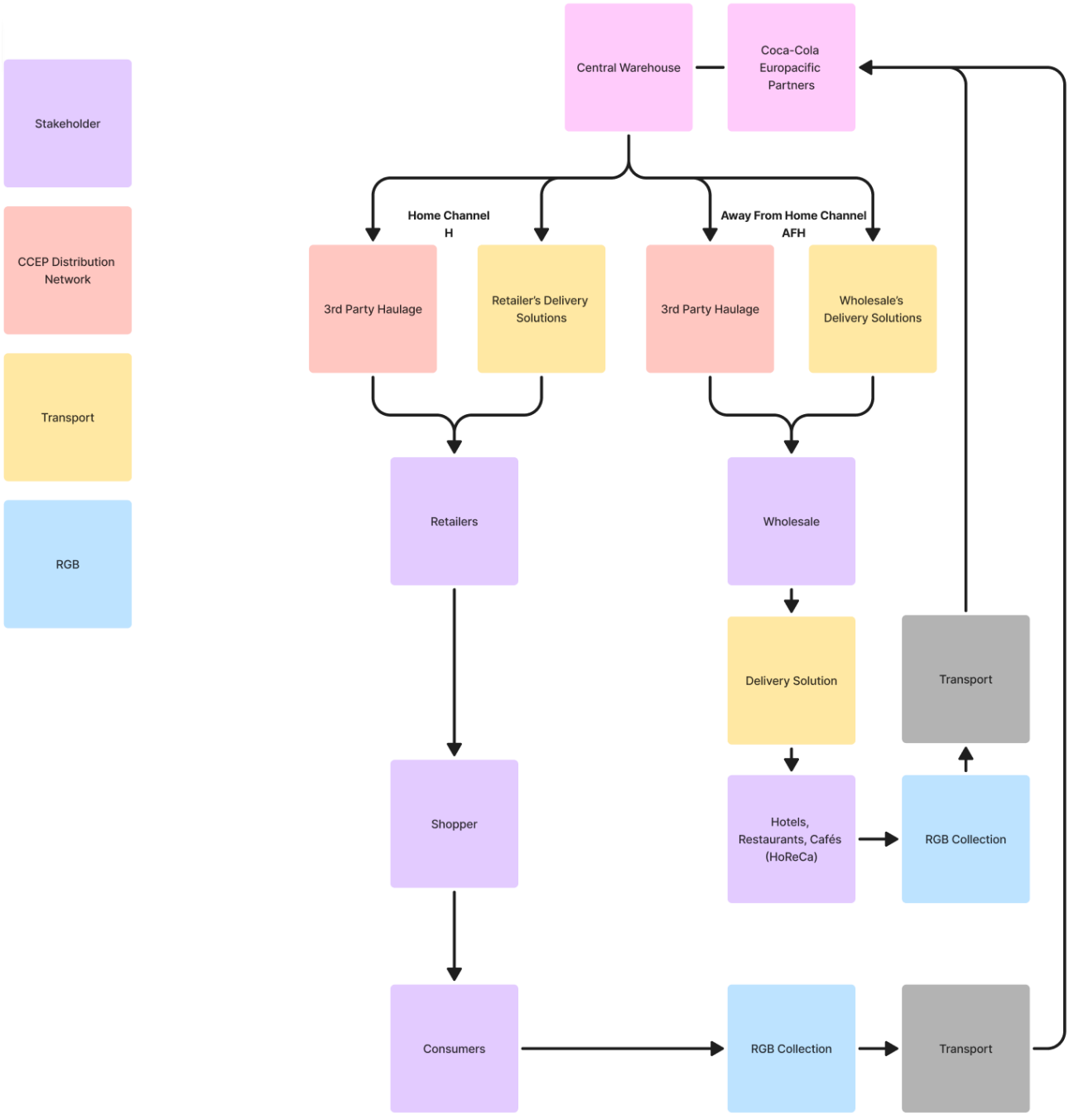


Figure D1. RGB Closed-Loop System

APPENDIX E. RISE LCA ASSUMPTIONS

This RISE LCA study relies on the following assumptions to evaluate the environmental impacts of glass bottle systems (Löfgren *et al.*, 2024).

Collection and Reuse

- **Collection Rate**: A 100% collection rate was assumed for refillable bottles to maximize reuse and minimize material loss.
- **Bottle Lifespan**: Each 33 cl RGB was assumed to complete 40 cycles, and each 50 cl bottle 8 cycles, based on Sveriges Bryggerier estimates.
- **Crate Use**: Return crates for 33 cl and 50 cl bottles were assumed to last 40 cycles (2 cycles/year over 20 years) and were made of recycled HDPE, as was the tray for 150 cl PET bottles.
- **Durability**: A 40-year lifespan was assumed for 33 cl RGBs and an 8-year lifespan for 50 cl RGBs.

Production and Materials

- **Production Data**: CO_2 emissions data from Ardagh's glass production applied to all glass bottles.
- **Recycled Content**: Recycled glass content ranged from 50% to 90%, with higher levels in colored glass.
- Brewery Processing
- **Process Contribution**: For RGBs 10% of total energy consumed at the bottlers were allocated for cleaning bottles.

Packaging and Pallets

- **Secondary Packaging**: Crates and shrink-wrap were excluded from direct calculations.
- **EUR Pallets**: EUR pallets, heavily reused, contributed minimally to emissions and were thus excluded, as in 2018. Stretch film was also excluded.

Transportation and Distribution

- **Distribution**: Filled bottles were transported using a fully loaded 22-ton diesel truck over a one-way distance of 250 km from distribution to bottler. Environmental impact was calculated based on the combined weight of the beverage, bottle, packaging, and pallet share.
- **Packaging material**: 500 km was assumed for transport distance for packaging material to bottler.
- **Return Transport**: RGBs were sent back for cleaning and refilling in the same truck,

Recycling and Waste Management

- **Recycling Process**: Collected glass was sent to SGÅ's Hammar facility for sorting, cleaning, and recycling, with diesel-powered transport data provided by SGÅ for 2022.
- **Plastic Crates and Cardboard**: RGB crates were recycled at a 33% rate, with the rest incinerated. Cardboard for single-use bottles was assumed to be recycled at 78% per FTI data.

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